ACTIVE PRELUDE TO CALCULUS

ACTIVITIES WORKBOOK – 2019 Edition

Matthew Boelkins

Activities Workbook
for
Active Prelude to Calculus

Activities Workbook for Active Prelude to Calculus

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August 6, 2024

Cover Photo: Noah Wyn Photography

Edition: 2019

Website: activecalculus.org/apc/

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Preface

This Activities Workbook for *Active Prelude to Calculus* collects all the Preview Activities and Activities in a way that each starts on a new page. The Activities Workbook is designed to be used by students who wish to have a complete set of the activities to work through as they read the book in an electronic format and for instructors who wish to have a one activity per page format to make printing for distribution in class easier.

The design of this workbook is such that each Preview Activity and Activity starts on a right-hand page. As a result, most left-hand pages in this workbook are intentionally left blank as a place for student work associated with one of the adjacent activities.

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CHAPTER 1

Relating Changing Quantities

1.1 Changing in Tandem

Preview Activity 1.1.1. Suppose that a rectangular aquarium is being filled with water. The tank is 4 feet long by 2 feet wide by 3 feet high, and the hose that is filling the tank is delivering water at a rate of 0.5 cubic feet per minute.

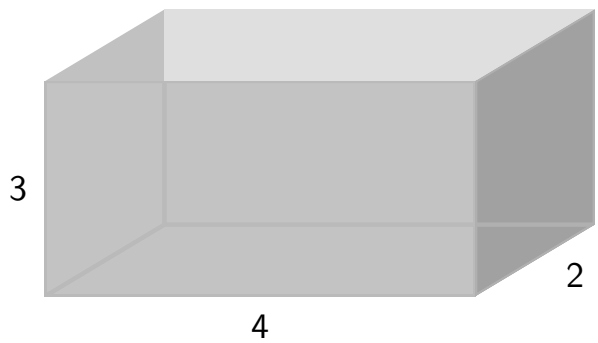


Figure 1.1.1 The empty aquarium.

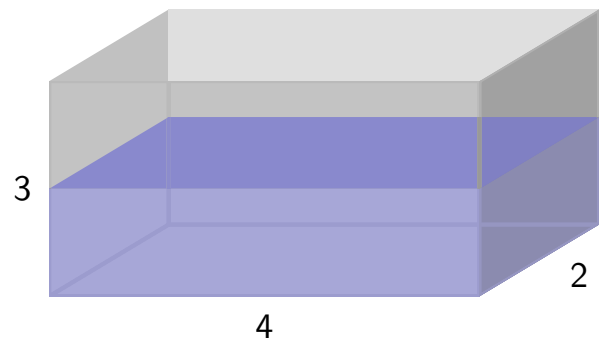


Figure 1.1.2 The aquarium, partially filled.

(a) What are some different quantities that are changing in this scenario?

(b) After 1 minute has elapsed, how much water is in the tank? At this moment, how deep is the water?

(c) How much water is in the tank and how deep is the water after 2 minutes? After 3 minutes?

(d) How long will it take for the tank to be completely full? Why?

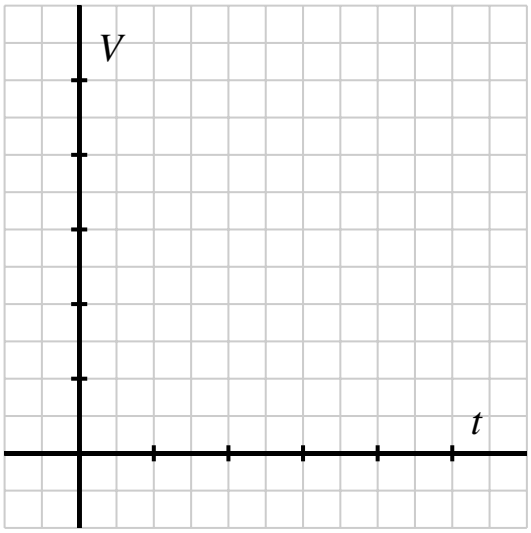
Activity 1.1.2. Consider a tank in the shape of an inverted circular cone (point down) where the tank's radius is 2 feet and its depth is 4 feet. Suppose that the tank is being filled with water that is entering at a constant rate of 0.75 cubic feet per minute.

(a) Sketch a labeled picture of the tank, including a snapshot of there being water in the tank prior to the tank being completely full.

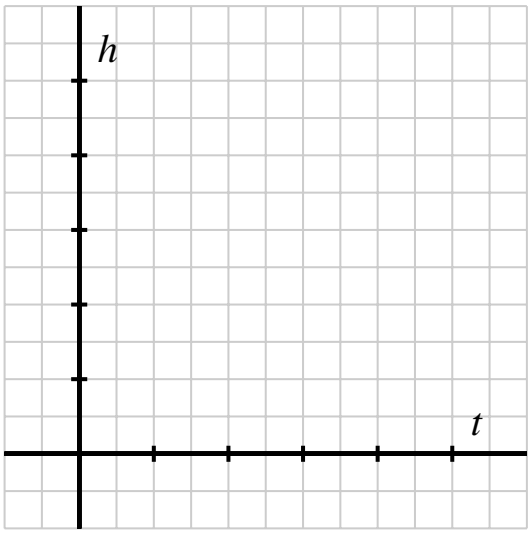
(b) What are some quantities that are changing in this scenario? What are some quantities that are not changing?

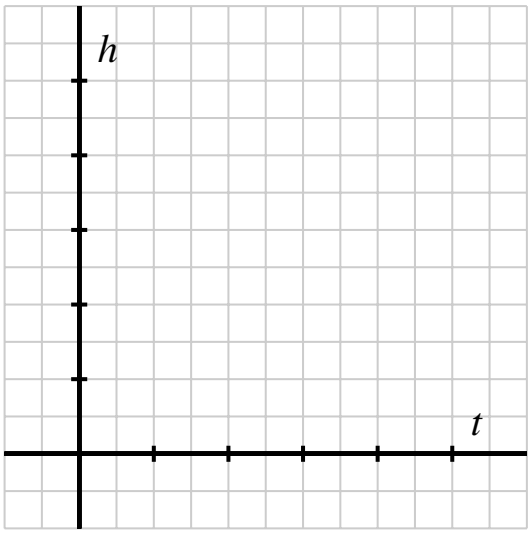
(c) Fill in the following table of values to determine how much water, V, is in the tank at a given time in minutes, t, and thus generate a graph of the relationship between volume and time by plotting the data on the provided axes.

t	V
0	
1	
2	
3	
4	
5	



(d) Finally, think about how the height, h, of the water changes in tandem with time. Without attempting to determine specific values of h at particular values of t, how would you expect the data for the relationship between h and t to appear? Use the provided axes to sketch at least two possibilities; write at least one sentence to explain how you think the graph should appear.





Activity 1.1.3. Consider a tank in the shape of a sphere where the tank's radius is 3 feet. Suppose that the tank is initially completely full and that it is being drained by a pump at a constant rate of 1.2 cubic feet per minute.

(a) Sketch a labeled picture of the tank, including a snapshot of some water remaining in the tank prior to the tank being completely empty.

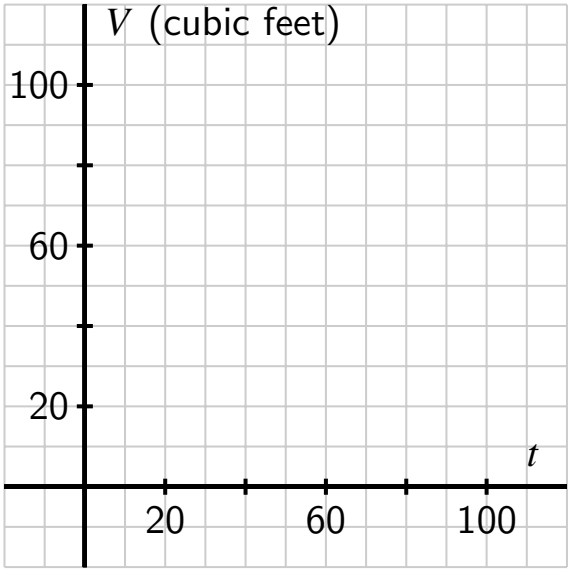
(b) What are some quantities that are changing in this scenario? What are some quantities that are not changing?

(c) Recall that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$. When the tank is completely full at time $t = 0$ right before it starts being drained, how much water is present?

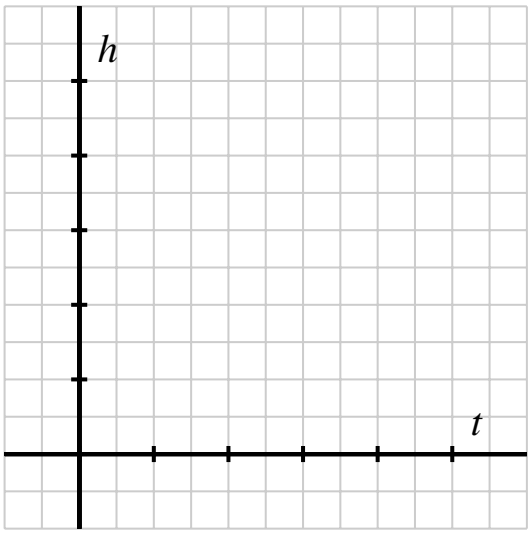
(d) How long will it take for the tank to drain completely?

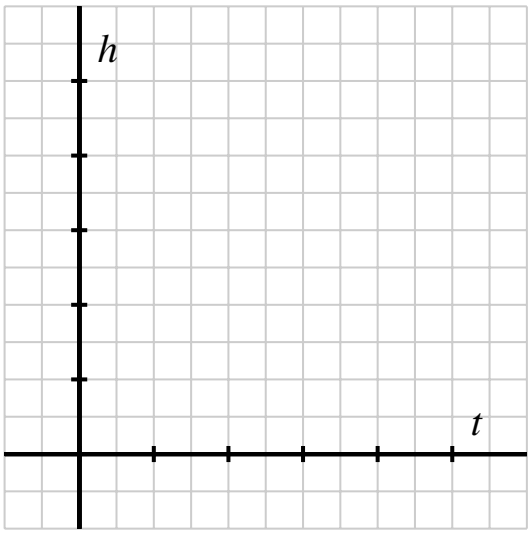
(e) Fill in the following table of values to determine how much water, V, is in the tank at a given time in minutes, t, and thus generate a graph of the relationship between volume and time. Write a sentence to explain why the data's graph appears the way that it does.

t	V
0	
20	
40	
60	
80	
94.24	



(f) Finally, think about how the height of the water changes in tandem with time. What is the height of the water when $t = 0$? What is the height when the tank is empty? How would you expect the data for the relationship between h and t to appear? Use the provided axes to sketch at least two possibilities; write at least one sentence to explain how you think the graph should appear.





1.2 Functions: Modeling Relationships

Preview Activity 1.2.1. Use the equation $T = 40 + 0.25N$ that relates the temperature, T, to the number of chirps per minute, N, to respond to the questions below. The equation is also called "Dolbear's Law".

(a) If we hear snowy tree crickets chirping at a rate of 92 chirps per minute, what does Dolbear's Law suggest should be the outside temperature?

(b) If the outside temperature is $77°$ F, how many chirps per minute should we expect to hear?

(c) Is the model valid for determining the number of chirps one should hear when the outside temperature is $35°$ F? Why or why not?

(d) Suppose that in the morning an observer hears 65 chirps per minute, and several hours later hears 75 chirps per minute. How much has the temperature risen between observations?

(e) Dolbear's Law is known to be accurate for temperatures from $50°$ to $85°$. What is the fewest number of chirps per minute an observer could expect to hear? the greatest number of chirps per minute?

Activity 1.2.2. Consider a spherical tank of radius 4 m that is filling with water. Let V be the volume of water in the tank (in cubic meters) at a given time, and h the depth of the water (in meters) at the same time. It can be shown using calculus that V is a function of h according to the rule

$$V = f(h) = \frac{\pi}{3}h^2(12 - h).$$

(a) What values of h make sense to consider in the context of this function? What values of V make sense in the same context?

(b) What is the domain of the function f in the context of the spherical tank? Why? What is the corresponding codomain? Why?

(c) Determine and interpret (with appropriate units) the values $f(2)$, $f(4)$, and $f(8)$. What is important about the value of $f(8)$?

(d) Consider the claim: "since $f(9) = \frac{\pi}{3}9^2(12 - 9) = 81\pi \approx 254.47$, when the water is 9 meters deep, there is about 254.47 cubic meters of water in the tank". Is this claim valid? Why or why not? Further, does it make sense to observe that "$f(13) = -\frac{169\pi}{3}$"? Why or why not?

(e) Can you determine a value of h for which $f(h) = 300$ cubic meters? Why or why not?

Activity 1.2.3. Consider a spherical tank of radius 4 m that is completely full of water. Suppose that the tank is being drained by regulating an exit valve in such a way that the height of the water in the tank is always decreasing at a rate of 0.5 meters per minute. Let V be the volume of water in the tank (in cubic meters) at a given time t (in minutes), and h the depth of the water (in meters) at the same time. It can be shown using calculus that V is a function of t according to the model

$$V = p(t) = \frac{256\pi}{3} - \frac{\pi}{24}t^2(24 - t).$$

In addition, let $h = q(t)$ be the function whose output is the depth of the water in the tank at time t.

(a) What is the height of the water when $t = 0$? When $t = 1$? When $t = 2$? How long will it take the tank to completely drain? Why?

(b) What is the domain of the model $h = q(t)$? What is the domain of the model $V = p(t)$?

(c) How much water is in the tank when the tank is full? What is the range of the model $h = q(t)$? What is the range of the model $V = p(t)$?

(d) We will frequently use a graphing utility to help us understand function behavior, and strongly recommend *Desmos* (`www.desmos.com/`) because it is intuitive, online, and free. To learn more about *Desmos*, see their outstanding online tutorials (`learn.desmos.com/`).

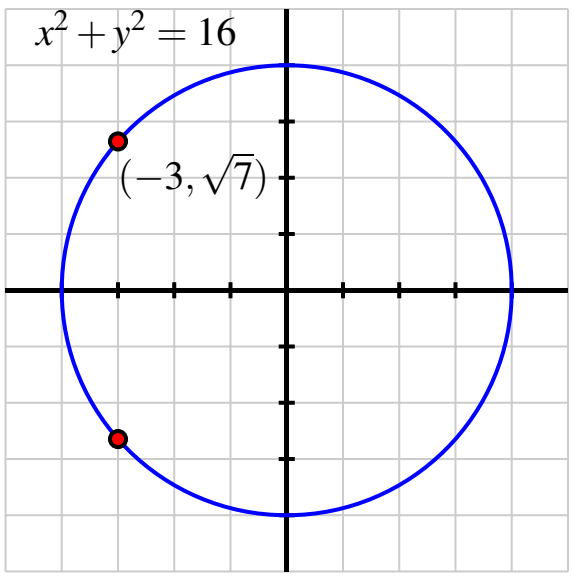
In this prepared *Desmos* worksheet (`www.desmos.com/calculator/btlyyxsdr4`), you can see how we enter the (abstract) function $V = p(t) = \frac{256\pi}{3} - \frac{\pi}{24}t^2(24 - t)$, as well as the corresponding graph the program generates. Make as many observations as you can about the model $V = p(t)$. You should discuss its shape and overall behavior, its domain, its range, and more.

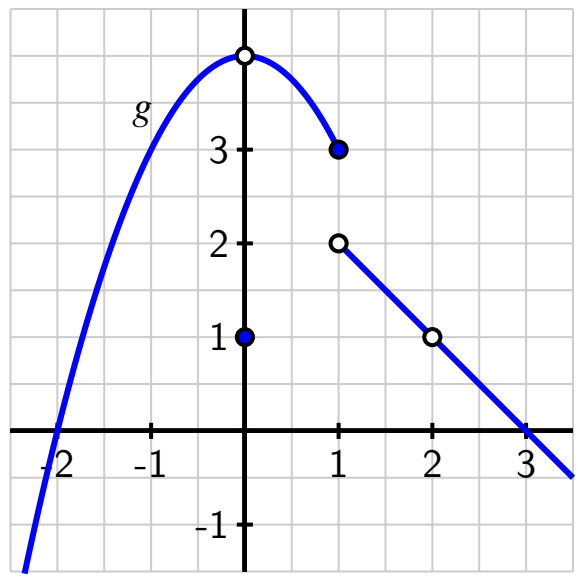
(e) How does the model $V = p(t) = \frac{256\pi}{3} - \frac{\pi}{24}t^2(24 - t)$ differ from the abstract function $y = r(x) = \frac{256\pi}{3} - \frac{\pi}{24}x^2(24 - x)$? In particular, how do the domain and range of the model differ from those of the abstract function, if at all?

(f) How should the graph of the height function $h = q(t)$ appear? Can you determine a formula for q? Explain your thinking.

Activity 1.2.4. Each of the following prompts describes a relationship between two quantities. For each, your task is to decide whether or not the relationship can be thought of as a function. If not, explain why. If so, state the domain and codomain of the function and write at least one sentence to explain the process that leads from the collection of inputs to the collection of outputs.

(a) The relationship between x and y in each of the graphs below (address each graph separately as a potential situation where y is a function of x). In the lefthand figure, any point on the circle relates x and y. For instance, the y-value $\sqrt{7}$ is related to the x-value -3. In the righthand figure, any point on the blue curve relates x and y. For instance, when $x = -1$, the corresponding y-value is $y = 3$. An unfilled circle indicates that there is not a point on the graph at that specific location.





(b) The relationship between the day of the year and the value of the S&P500 stock index (at the close of trading on a given day), where we attempt to consider the index's value (at the close of trading) as a function of the day of the year.

(c) The relationship between a car's velocity and its odometer, where we attempt to view the car's odometer reading as a function of its velocity.

(d) The relationship between x and y that is given in the following table where we attempt to view y as depending on x.

x	1	2	3	2	1
y	11	12	13	14	15

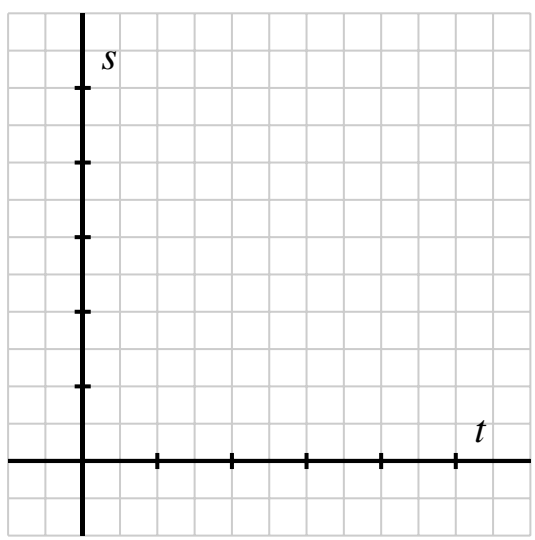
1.3 The Average Rate of Change of a Function

Preview Activity 1.3.1. Let the height function for a ball tossed vertically be given by $s(t) = 64 - 16(t-1)^2$, where t is measured in seconds and s is measured in feet above the ground.

(a) Compute the value of $AV_{[1.5,2.5]}$.

(b) What are the units on the quantity $AV_{[1.5,2.5]}$? What is the meaning of this number in the context of the rising/falling ball?

(c) In *Desmos*, plot the function $s(t) = 64-16(t-1)^2$ along with the points $(1.5, s(1.5))$ and $(2.5, s(2.5))$. Make a copy of your plot on the axes in the figure provided, labeling key points as well as the scale on your axes. What is the domain of the model? The range? Why?



(d) Work by hand to find the equation of the line through the points $(1.5, s(1.5))$ and $(2.5, s(2.5))$. Write the line in the form $y = mt + b$ and plot the line in *Desmos*, as well as on the axes above.

(e) What is a geometric interpretation of the value $AV_{[1.5,2.5]}$ in light of your work in the preceding questions?

(f) How do your answers in the preceding questions change if we instead consider the interval $[0.25, 0.75]$? $[0.5, 1.5]$? $[1, 3]$?

Activity 1.3.2. According to the US census, the populations of Kent and Ottawa Counties in Michigan where GVSU is located (Grand Rapids is in Kent, Allendale in Ottawa) from 1960 to 2010 measured in 10-year intervals are given in the following tables.

Year:	1960	1970	1980	1990	2000	2010
Kent County population:	363,187	411,044	444,506	500,631	574,336	602,622

Year:	1960	1970	1980	1990	2000	2010
Ottawa County Population:	98,719	128,181	157,174	187,768	238,313	263,801

Let $K(Y)$ represent the population of Kent County in year Y and $W(Y)$ the population of Ottawa County in year Y.

(a) Compute $AV_{[1990,2010]}$ for both K and W.

(b) What are the units on each of the quantities you computed in (a.)?

(c) Write a careful sentence that explains the meaning of the average rate of change of the Ottawa county population on the time interval $[1990, 2010]$. Your sentence should begin something like "In an average year between 1990 and 2010, the population of Ottawa County was . . ."

(d) Which county had a greater average rate of change during the time interval $[2000, 2010]$? Were there any intervals in which one of the counties had a negative average rate of change?

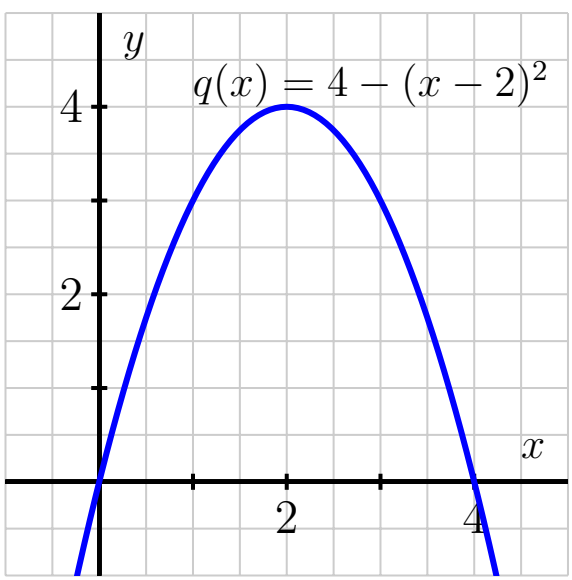
(e) Using the given data, what do you predict will be the population of Ottawa County in 2018? Why?

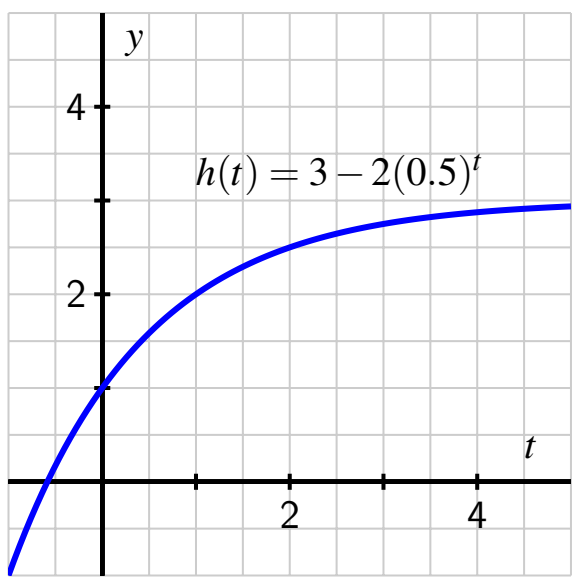
Activity 1.3.3. Let's consider two different functions and see how different computations of their average rate of change tells us about their respective behavior. Plots of q and h are shown in the figures in part (c).

(a) Consider the function $q(x) = 4 - (x-2)^2$. Compute $AV_{[0,1]}$, $AV_{[1,2]}$, $AV_{[2,3]}$, and $AV_{[3,4]}$. What do your last two computations tell you about the behavior of the function q on $[2, 4]$?

(b) Consider the function $h(t) = 3 - 2(0.5)^t$. Compute $AV_{[-1,1]}$, $AV_{[1,3]}$, and $AV_{[3,5]}$. What do your computations tell you about the behavior of the function h on $[-1, 5]$?

(c) On the graphs that follow (q at left, h at right), plot the line segments whose respective slopes are the average rates of change you computed in (a) and (b).



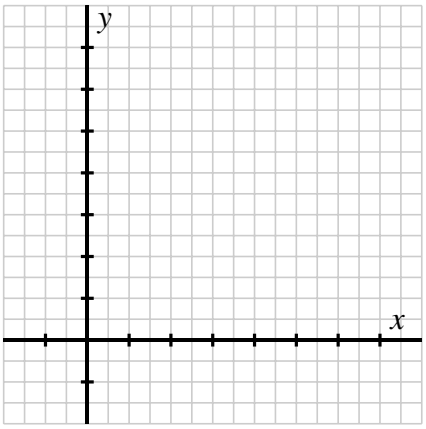


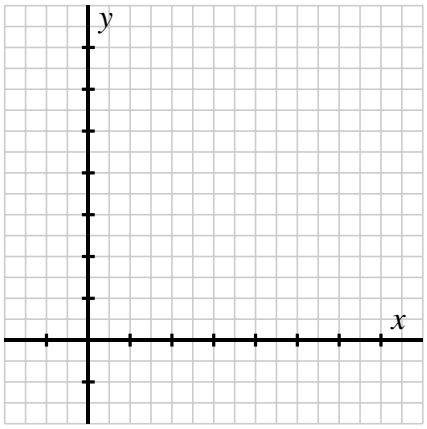
(d) True or false: Since $AV_{[0,3]} = 1$, the function q is increasing on the interval $(0, 3)$. Justify your decision.

(e) Give an example of a function that has the same average rate of change no matter what interval you choose. You can provide your example through a table, a graph, or a formula; regardless of your choice, write a sentence to explain.

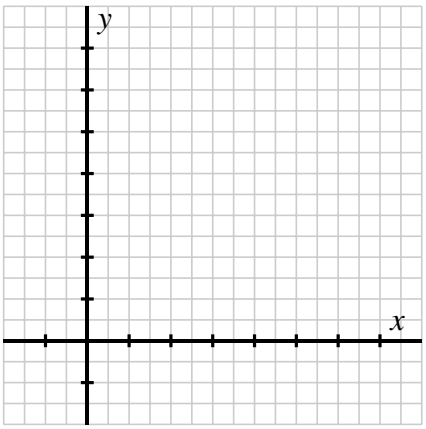
Activity 1.3.4. Sketch at least two different possible graphs that satisfy the criteria for the function stated in each part. Make your graphs as significantly different as you can. If it is impossible for a graph to satisfy the criteria, explain why.

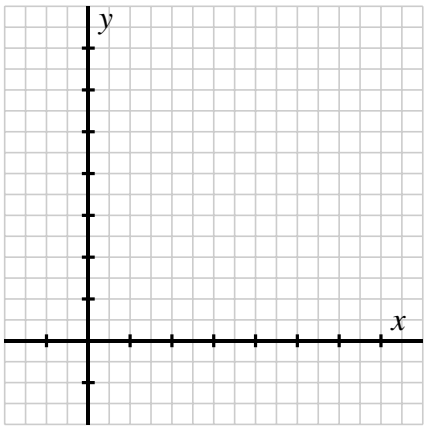
(a) f is a function defined on $[-1, 7]$ such that $f(1) = 4$ and $AV_{[1,3]} = -2$.



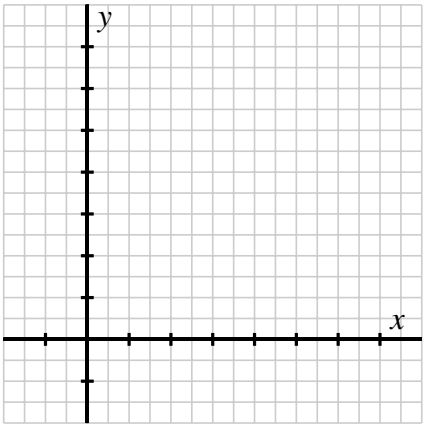


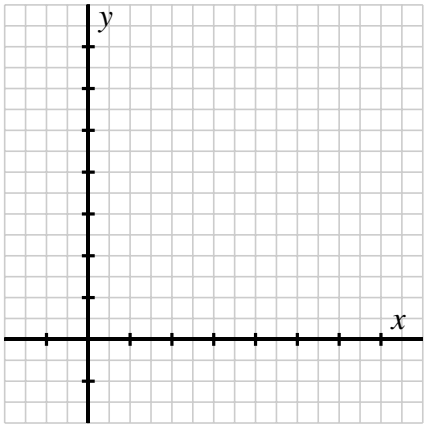
(b) g is a function defined on $[-1, 7]$ such that $g(4) = 3$, $AV_{[0,4]} = 0.5$, and g is not always increasing on $(0, 4)$.





(c) h is a function defined on $[-1, 7]$ such that $h(2) = 5$, $h(4) = 3$ and $AV_{[2,4]} = -2$.





1.4 Linear Functions

Preview Activity 1.4.1.

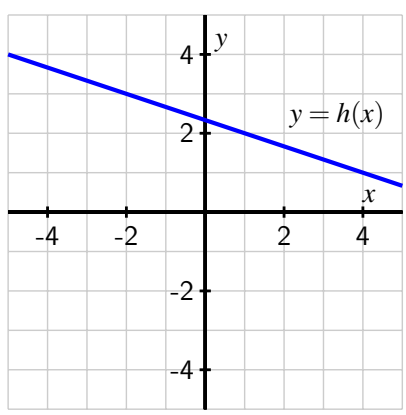
(a) Let $y = f(x) = 7 - 3x$. Determine $AV_{[-3,-1]}$, $AV_{[2,5]}$, and $AV_{[4,10]}$ for the function f.

(b) Let $y = g(x)$ be given by the data in the following table.

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
$g(x)$	-2.75	-2.25	-1.75	-1.25	-0.75	-0.25	0.25	0.75	1.25	1.75	2.25

Determine $AV_{[-5,-2]}$, $AV_{[-1,1]}$, and $AV_{[0,4]}$ for the function g.

(c) Consider the function $y = h(x)$ defined by the graph in the following figure.



Determine $AV_{[-5,-2]}$, $AV_{[-1,1]}$, and $AV_{[0,4]}$ for the function h.

(d) What do all three examples above have in common? How do they differ?

(e) For the function $y = f(x) = 7 - 3x$ from (a), find the simplest expression you can for

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

where $a \neq b$.

Activity 1.4.2. Find an equation for the line that is determined by the following conditions; write your answer in point-slope form wherever possible.

(a) The line with slope $\frac{3}{7}$ that passes through $(-11, -17)$.

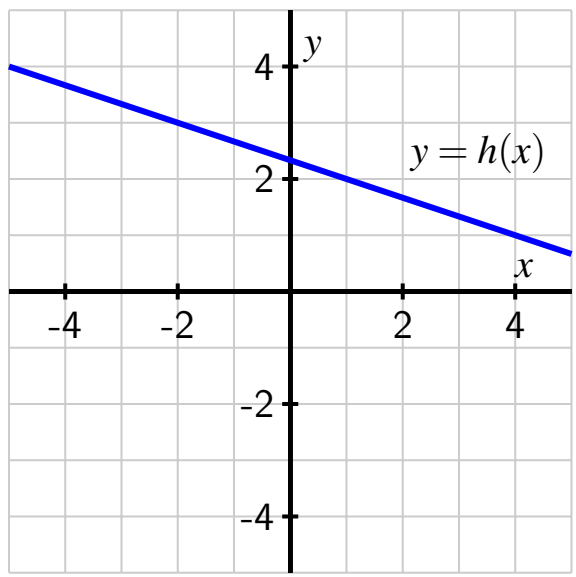
(b) The line passing through the points $(-2, 5)$ and $(3, -1)$.

(c) The line passing through $(4, 9)$ that is parallel to the line $2x - 3y = 5$.

(d) Explain why the function f given by the data in the following table appears to be linear and find a formula for $f(x)$.

x	1	3	4	7
$f(x)$	7	3	1	-5

(e) Find a formula for the linear function shown in the following figure.



Activity 1.4.3. The summit of Africa's largest peak, Mt. Kilimanjaro, has two main ice fields and a glacier at its peak. Geologists measured the ice cover in the year 2000 ($t = 0$) to be approximately 1951 m^2; in the year 2007, the ice cover measured 1555 m^2.

(a) Suppose that the amount of ice cover at the peak of Mt. Kilimanjaro is changing at a constant average rate from year to year. Find a linear model $A = f(t)$ whose output is the area of the ice cover, A, in square meters in year t (where t is the number of years after 2000).

(b) What do the slope and A-intercept mean in the model you found in (a)? In particular, what are the units on the slope?

(c) Compute $f(17)$. What does this quantity measure? Write a complete sentence to explain.

(d) If the model holds further into the future, when do we predict the ice cover will vanish?

(e) In light of your work above, what is a reasonable domain to use for the model $A = f(t)$? What is the corresponding range?

The main context of the sequence of questions in this activity comes from Exercise 30 on p. 27 of Connally's *Functions Modeling Change*, 5th ed.

Activity 1.4.4. In each of the following prompts, we investigate linear functions in context.

(a) A town's population initially has 28750 people present and then grows at a constant rate of 825 people per year. Find a linear model $P = f(t)$ for the number of people in the town in year t.

(b) A different town's population Q is given by the function $Q = g(t) = 42505 - 465t$. What is the slope of this function and what is its meaning in the model? Write a complete sentence to explain.

(c) A spherical tank is being drained with a pump. Initially the tank is full with $\frac{32\pi}{3}$ cubic feet of water. Assume the tank is drained at a constant rate of 1.2 cubic feet per minute. Find a linear model $V = p(t)$ for the total amount of water in the tank at time t. In addition, what is a reasonable approximate domain for the model?

(d) A conical tank is being filled in such a way that the height of the water in the tank, h (in feet), at time t (in minutes) is given by the function $h = q(t) = 0.65t$. What can you say about how the water level is rising? Write at least one careful sentence to explain.

(e) Suppose we know that a 5-year old car's value is \$10200, and that after 10 years its value is \$4600. Assuming that the car's value depreciates linearly, find a function $C = L(t)$ whose output is the value of the car in year t. What is a reasonable domain for the model? What is the value and meaning of the slope of the line? Write at least one careful sentence to explain.

1.5 Quadratic Functions

Preview Activity 1.5.1. A water balloon is tossed vertically from a fifth story window. Its height, h, in meters, at time t, in seconds, is modeled by the function

$$h = q(t) = -5t^2 + 20t + 25.$$

(a) Execute appropriate computations to complete both of the following tables: values of the function h on the left, average rates of change for h on the right.

t	$h = q(t)$
0	$q(0) = 25$
1	
2	
3	
4	
5	

$[a, b]$	$AV_{[a,b]}$
$[0, 1]$	$AV_{[0,1]} = 15$ m/s
$[1, 2]$	
$[2, 3]$	
$[3, 4]$	
$[4, 5]$	

(b) What pattern(s) do you observe in the table of function values and in the table of average rates of change?

(c) Explain why $h = q(t)$ is not a linear function. Use the definition of a linear function (that is, referencing average rate of change) in your response.

(d) What is the average velocity of the water balloon in the final second before it lands? How does this value compare to the average velocity on the time interval $[4.9, 5]$?

Activity 1.5.2. Open a browser and point it to *Desmos* (`www.desmos.com/calculator`). In *Desmos*, enter `q(x) = ax^2 + bx + c`; you will be prompted to add sliders for a, b, and c. Do so. Then begin exploring with the sliders and respond to the following questions.

(a) Describe how changing the value of a affects the graph of q.

(b) Describe how changing the value of b affects the graph of q.

(c) Describe how changing the value of c affects the graph of q.

(d) Which parameter seems to have the simplest effect? Which parameter seems to have the most complicated effect? Why?

(e) Is it possible to find a formula for a quadratic function that passes through the points $(0, 8)$, $(1, 12)$, $(2, 12)$? If yes, do so; if not, explain why not.

Activity 1.5.3. Reason algebraically using appropriate properties of quadratic functions to answer the following questions. Use *Desmos* to check your results graphically.

(a) How many quadratic functions have x-intercepts at $(-5, 0)$ and $(10, 0)$ and a y-intercept at $(0, -1)$? Can you determine an exact formula for such a function? If yes, do so. If not, explain why.

(b) Suppose that a quadratic function q has vertex $(-3, -4)$ and opens upward. How many x-intercepts can you guarantee the function has? Why?

(c) In addition to the information in (b), suppose you know that $q(-1) = -3$. Can you determine an exact formula for q? If yes, do so. If not, explain why.

(d) Does the quadratic function $p(x) = -3(x+1)^2 + 9$ have 0, 1, or 2 x-intercepts? Reason algebraically to determine the exact values of any such intercepts or explain why none exist.

(e) Does the quadratic function $w(x) = -2x^2 + 10x - 20$ have 0, 1, or 2 x-intercepts? Reason algebraically to determine the exact values of any such intercepts or explain why none exist.

Activity 1.5.4. A water balloon is tossed vertically from a window at an initial height of 37 feet and with an initial velocity of 41 feet per second.

(a) Determine a formula, $s(t)$, for the function that models the height of the water balloon at time t.

(b) Plot the function in *Desmos* in an appropriate window. Sketch a copy of the graph here.

(c) Use the graph to estimate the time the water balloon lands.

(d) Use algebra to find the exact time the water balloon lands.

(e) Determine the exact time the water balloon reaches its highest point and its height at that time.

(f) Compute the average rate of change of s on the intervals $[1.5, 2]$, $[2, 2.5]$, $[2.5, 3]$. Include units on your answers and write one sentence to explain the meaning of the values you found. Sketch appropriate lines on the graph of s whose respective slopes are the values of these average rates of change.

1.6 Composite Functions

Preview Activity 1.6.1. Let $y = p(x) = 3x - 4$ and $x = q(t) = t^2 - 1$.

(a) Let $r(t) = p(q(t))$. Determine a formula for r that depends only on t and not on p or q.

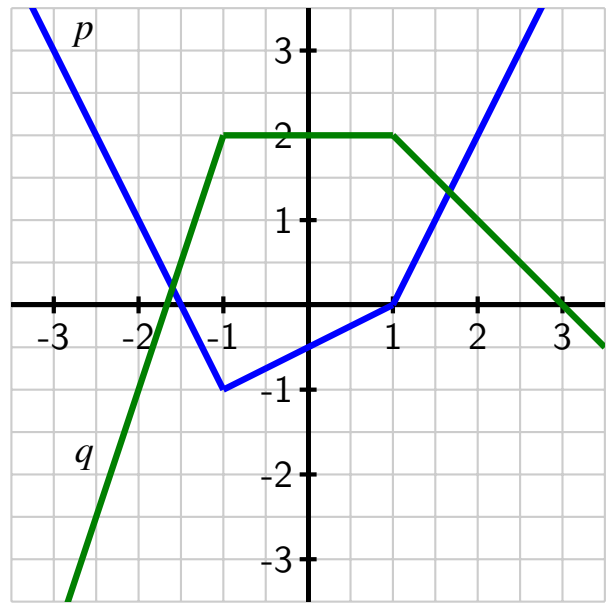
(b) Review the introductory example with $f(x) = x^2 - 1$ and $g(t) = 3t - 4$, which involved functions similar to p and q in part (a). What is the biggest difference between your work in (a) above and in the introductory example?

(c) Let $t = s(z) = \frac{1}{z+4}$ and recall that $x = q(t) = t^2 - 1$. Determine a formula for $x = q(s(z))$ that depends only on z.

(d) Suppose that $h(t) = \sqrt{2t^2 + 5}$. Determine formulas for two related functions, $y = f(x)$ and $x = g(t)$, so that $h(t) = f(g(t))$.

Activity 1.6.2. Let functions p and q be given by the graphs in the figure below, which are each piecewise linear - that is, parts that look like straight lines are straight lines. In addition, let f and g be given by the tables below.

x	0	1	2	3	4
$f(x)$	6	4	3	4	6
$g(x)$	1	3	0	4	2



Compute each of the following quantities or explain why they are not defined.

(a) $p(q(0))$

(b) $q(p(0))$

(c) $(p \circ p)(-1)$

(d) $(f \circ g)(2)$

(e) $(g \circ f)(3)$

(f) $g(f(0))$

(g) For what value(s) of x is $f(g(x)) = 4$?

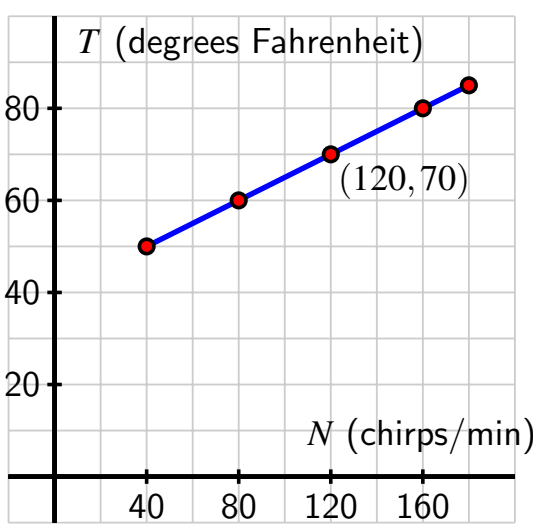
(h) For what value(s) of x is $q(p(x)) = 1$?

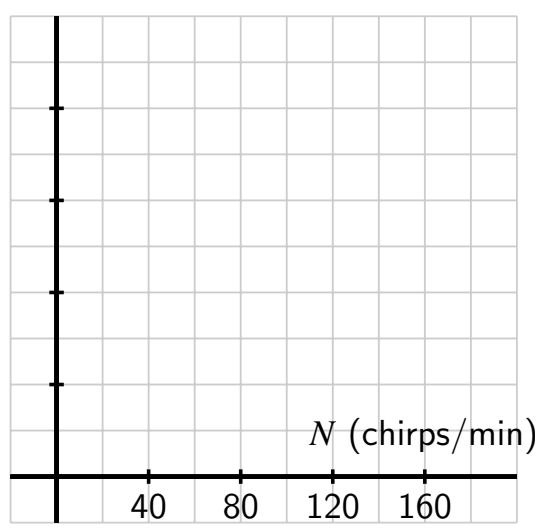
Activity 1.6.3. Let $F = D(N) = 40 + 0.25N$ be Dolbear's function that converts an input of number of chirps per minute to degrees Fahrenheit, and let $C = G(F) = \frac{5}{9}(F - 32)$ be the function that converts an input of degrees Fahrenheit to an output of degrees Celsius.

(a) Determine a formula for the new function $H = (G \circ D)$ that depends only on the variable N.

(b) What is the meaning of the function you found in (a)?

(c) How does a plot of the function $H = (G \circ D)$ compare to that of Dolbear's function? Sketch a plot of $y = H(N) = (G \circ D)(N)$ on the blank axes to the right of the plot of Dolbear's function, and discuss the similarities and differences between them. Be sure to label the vertical scale on your axes.





(d) What is the domain of the function $H = G \circ D$? What is its range?

Activity 1.6.4. Let $f(x) = 2x^2 - 3x + 1$ and $g(x) = \frac{5}{x}$.

(a) Compute $f(1 + h)$ and expand and simplify the result as much as possible by combining like terms.

(b) Determine the most simplified expression you can for the average rate of change of f on the interval $[1, 1 + h]$. That is, determine $AV_{[1,1+h]}$ for f and simplify the result as much as possible.

(c) Compute $g(1 + h)$. Is there any valid algebra you can do to write $g(1 + h)$ more simply?

(d) Determine the most simplified expression you can for the average rate of change of g on the interval $[1, 1 + h]$. That is, determine $AV_{[1,1+h]}$ for g and simplify the result.

1.7 Inverse Functions

Preview Activity 1.7.1. Recall that $F = g(C) = \frac{9}{5}C + 32$ is the function that takes Celsius temperature inputs and produces the corresponding Fahrenheit temperature outputs.

(a) Show that it is possible to solve the equation $F = \frac{9}{5}C + 32$ for C in terms of F and that doing so results in the equation $C = \frac{5}{9}(F - 32)$.

(b) Note that the equation $C = \frac{5}{9}(F - 32)$ expresses C as a function of F. Call this function h so that $C = h(F) = \frac{5}{9}(F - 32)$.

Find the simplest expression that you can for the composite function $j(C) = h(g(C))$.

(c) Find the simplest expression that you can for the composite function $k(F) = g(h(F))$.

(d) Why are the functions j and k so simple? Explain by discussing how the functions g and h process inputs to generate outputs and what happens when we first execute one followed by the other.

Activity 1.7.2. Recall Dolbear's function $F = D(N) = 40 + \frac{1}{4}N$ that converts the number, N, of snowy tree cricket chirps per minute to a corresponding Fahrenheit temperature. We have earlier established that the domain of D is $[40, 180]$ and the range of D is $[50, 85]$.

(a) Solve the equation $F = 40 + \frac{1}{4}N$ for N in terms of F. Call the resulting function $N = E(F)$.

(b) Explain in words the process or effect of the function $N = E(F)$. What does it take as input? What does it generate as output?

(c) Use the function E that you found in (a.) to compute $j(N) = E(D(N))$. Simplify your result as much as possible. Do likewise for $k(F) = D(E(F))$. What do you notice about these two composite functions j and k?

(d) Consider the equations $F = 40 + \frac{1}{4}N$ and $N = 4(F - 40)$. Do these equations express different relationships between F and N, or do they express the same relationship in two different ways? Explain.

Activity 1.7.3. Determine, with justification, whether each of the following functions has an inverse function. For each function that has an inverse function, give two examples of values of the inverse function by writing statements such as "$s^{-1}(3) = 1$".

(a) The function $f : S \to S$ given by the table of values below, where $S = \{0, 1, 2, 3, 4\}$.

x	0	1	2	3	4
$f(x)$	1	2	4	3	2

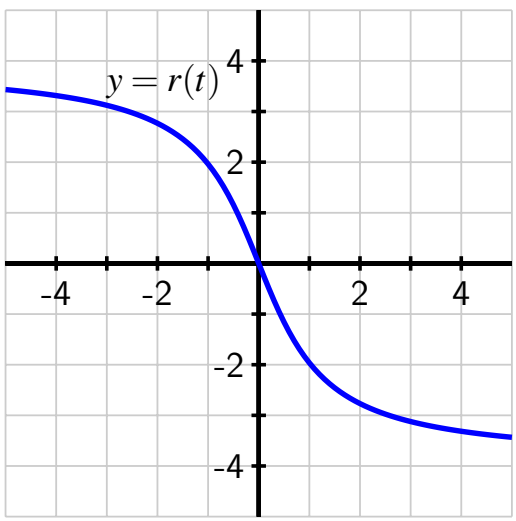
(b) The function $g : S \to S$ given by the table of values below, where $S = \{0, 1, 2, 3, 4\}$.

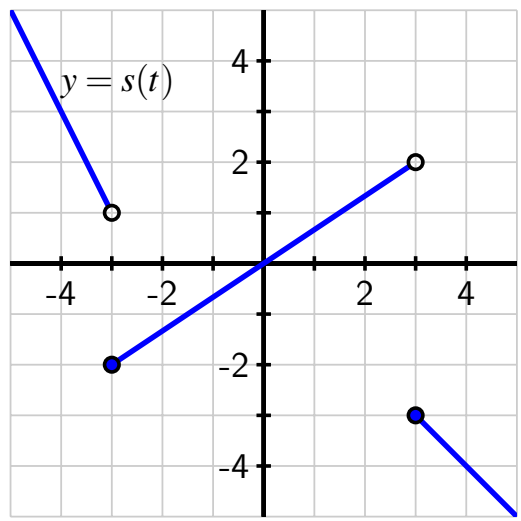
x	0	1	2	3	4
$g(x)$	4	0	3	1	2

(c) The function p given by $p(t) = 7 - \frac{3}{5}t$. Assume that the domain and codomain of p are both "all real numbers".

(d) The function q given by $q(t) = 7 - \frac{3}{5}t^4$. Assume that the domain and codomain of q are both "all real numbers".

(e) The functions r and s given by the graphs in the figures below. Assume that the graphs show all of the important behavior of the functions and that the apparent trends continue beyond what is pictured.





Activity 1.7.4. During a major rainstorm, the rainfall at Gerald R. Ford Airport is measured on a frequent basis for a 10-hour period of time. The following function g models the *rate*, R, at which the rain falls (in cm/hr) on the time interval $t = 0$ to $t = 10$:

$$R = g(t) = \frac{4}{t+2} + 1.$$

(a) Compute $g(3)$ and write a complete sentence to explain its meaning in the given context, including units.

(b) Compute the average rate of change of g on the time interval $[3, 5]$ and write two careful complete sentences to explain the meaning of this value in the context of the problem, including units. Explicitly address what the value you compute tells you about how rain is falling over a certain time interval, and what you should expect as time goes on.

(c) Plot the function $y = g(t)$ using a computational device. On the domain $[0, 10]$, what is the corresponding range of g? Why does the function g have an inverse function?

(d) Determine $g^{-1}\left(\frac{9}{5}\right)$ and write a complete sentence to explain its meaning in the given context.

(e) According to the model g, is there ever a time during the storm that the rain falls at a rate of exactly 1 centimeter per hour? Why or why not? Provide an algebraic justification for your answer.

1.8 Transformations of Functions

Preview Activity 1.8.1. Open a new *Desmos* graph and define the function $f(x) = x^2$. Adjust the window so that the range is for $-4 \leq x \leq 4$ and $-10 \leq y \leq 10$.

(a) In *Desmos*, define the function $g(x) = f(x) + a$. (That is, in *Desmos* on line 2, enter `g(x) = f(x) + a`.) You will get prompted to add a slider for a. Do so.

Explore by moving the slider for a and write at least one sentence to describe the effect that changing the value of a has on the graph of g.

(b) Next, define the function $h(x) = f(x - b)$. (That is, in *Desmos* on line 4, enter `h(x) = f(x-b)` and add the slider for b.)

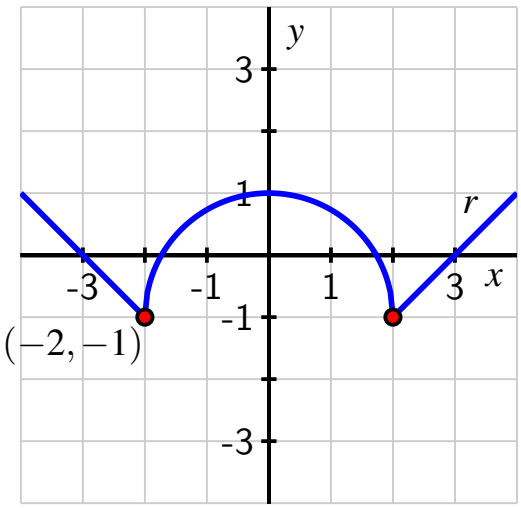
Move the slider for b and write at least one sentence to describe the effect that changing the value of b has on the graph of h.

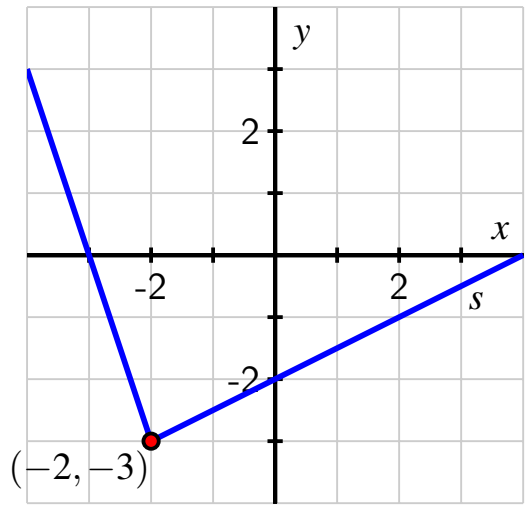
(c) Now define the function $p(x) = cf(x)$. (That is, in *Desmos* on line 6, enter `p(x) = cf(x)` and add the slider for c.)

Move the slider for c and write at least one sentence to describe the effect that changing the value of c has on the graph of p. In particular, when $c = -1$, how is the graph of p related to the graph of f?

(d) Finally, click on the icons next to g, h, and p to temporarily hide them, and go back to Line 1 and change your formula for f. You can make it whatever you'd like, but try something like $f(x) = x^2 + 2x + 3$ or $f(x) = x^3 - 1$. Then, investigate with the sliders a, b, and c to see the effects on g, h, and p (unhiding them appropriately). Write a couple of sentences to describe your observations of your explorations.

Activity 1.8.2. Consider the functions r and s given in the figures below.



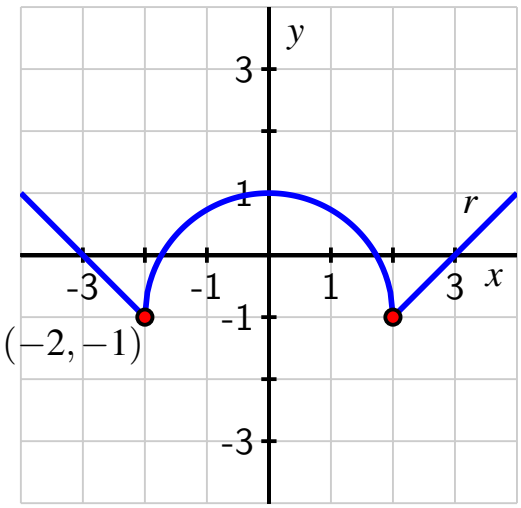


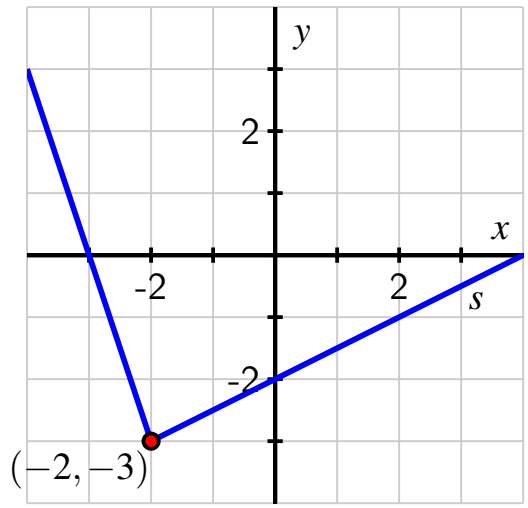
(a) On the same axes as the plot of $y = r(x)$, sketch the following graphs: $y = g(x) = r(x) + 2$, $y = h(x) = r(x + 1)$, and $y = f(x) = r(x + 1) + 2$. Be sure to label the point on each of g, h, and f that corresponds to $(-2, -1)$ on the original graph of r. In addition, write one sentence to explain the overall transformations that have resulted in g, h, and f.

(b) On the same axes as the plot of $y = s(x)$, sketch the following graphs: $y = k(x) = s(x) - 1$, $y = j(x) = s(x - 2)$, and $y = m(x) = s(x - 2) - 1$. Be sure to label the point on each of k, j, and m that corresponds to $(-2, -3)$ on the original graph of s. In addition, write one sentence to explain the overall transformations that have resulted in k, j, and m.

(c) Now consider the function $q(x) = x^2$. Determine a formula for the function that is given by $p(x) = q(x+3) - 4$. How is p a transformation of q?

Activity 1.8.3. Consider the functions r and s given in the figures below.

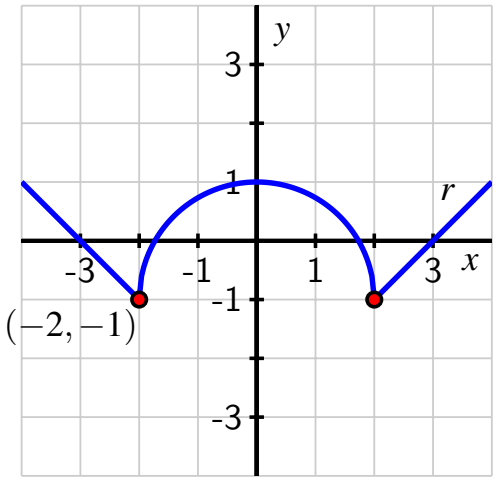


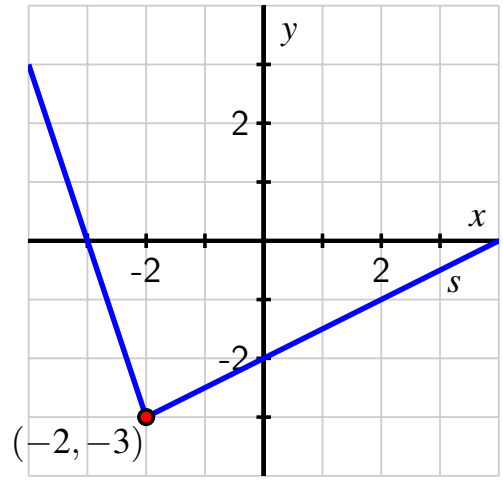


(a) On the same axes as the plot of $y = r(x)$, sketch the following graphs: $y = g(x) = 3r(x)$ and $y = h(x) = \frac{1}{3}r(x)$. Be sure to label the point on g and h that corresponds to the point $(-2, -1)$ on the original graph of r. In addition, write one sentence to explain the overall transformations that have resulted in g and h from r.

(b) On the same axes as the plot of $y = s(x)$, sketch the following graphs: $y = k(x) = -s(x)$ and $y = j(x) = -\frac{1}{2}s(x)$. Be sure to label the point on k and j that corresponds to the point $(-2, -3)$ on the original graph of s. In addition, write one sentence to explain the overall transformations that have resulted in k and j from s.

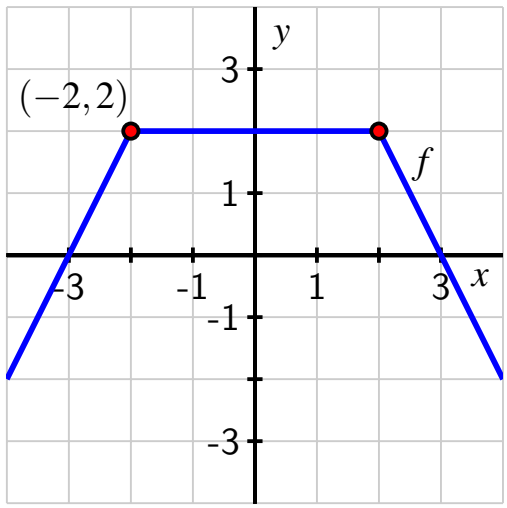
(c) On the additional copies of the two figures below, sketch the graphs of the following transformed functions: $y = m(x) = 2r(x+1) - 1$ (at left) and $y = n(x) = \frac{1}{2}s(x-2) + 2$. As above, be sure to label a key point on each graph that corresonds to the labeled point on the original parent function.

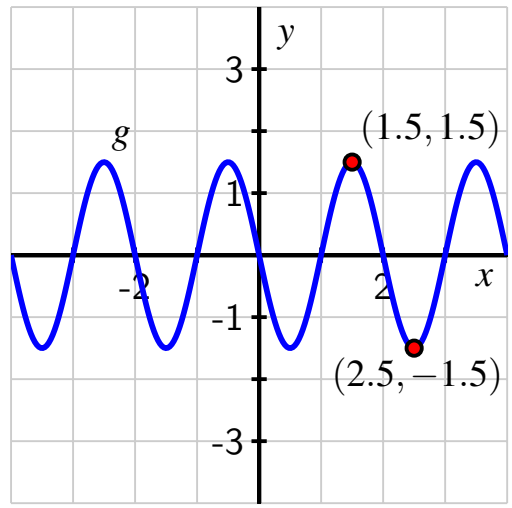




(d) Describe in words how the function $y = m(x) = 2r(x+1) - 1$ is the result of three elementary transformations of $y = r(x)$. Does the order in which these transformations occur matter? Why or why not?

Activity 1.8.4. Consider the functions f and g given in the following figures.



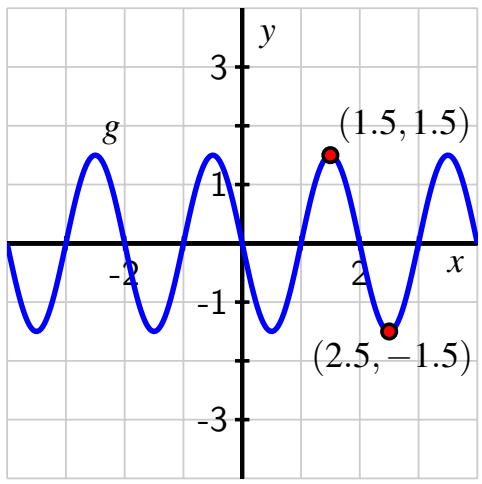


(a) Sketch an accurate graph of the transformation $y = p(x) = -\frac{1}{2}f(x-1)+2$. Write at least one sentence to explain how you developed the graph of p, and identify the point on p that corresponds to the original point $(-2, 2)$ on the graph of f.

(b) Sketch an accurate graph of the transformation $y = q(x) = 2g(x+0.5)-0.75$. Write at least one sentence to explain how you developed the graph of q, and identify the point on q that corresponds to the original point $(1.5, 1.5)$ on the graph of g.

(c) Is the function $y = r(x) = \frac{1}{2}(-f(x-1)-4)$ the same function as p in part (a) or different? Why? Explain in two different ways: discuss the algebraic similarities and differences between p and r, and also discuss how each is a transformation of f.

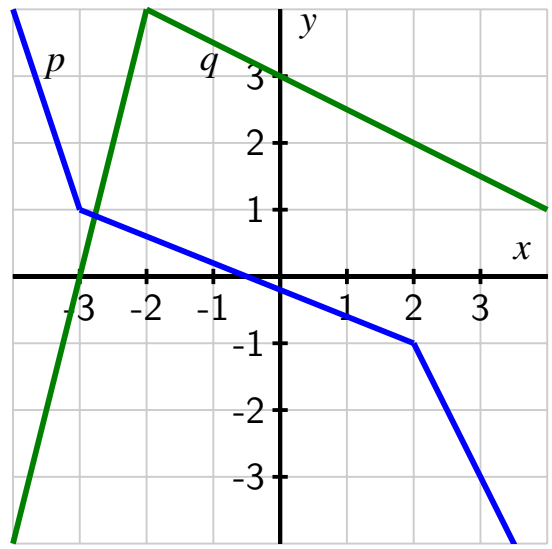
(d) Find a formula for a function $y = s(x)$ (in terms of g) that represents this transformation of g: a horizontal shift of 1.25 units left, followed by a reflection across the x-axis and a vertical stretch by a factor of 2.5 units, followed by a vertical shift of 1.75 units. Sketch an accurate, labeled graph of s on the following axes along with the given parent function g.



1.9 Combining Functions

Preview Activity 1.9.1. Consider the functions f and g defined by the following table and the piecewise linear functions p and q defined by the following figure. Assume that the lines in the figure pass through whole number coordinates where they appear to do so; for example, $(2, 2)$ lies on the graph of q, and $(3, -3)$ lies on the graph of p.

x	0	1	2	3	4
$f(x)$	5	10	15	20	25
$g(x)$	9	5	3	2	3



(a) Let $h(x) = f(x) + g(x)$. Determine $h(3)$.

(b) Let $r(x) = p(x) - q(x)$. Determine $r(-1)$ exactly.

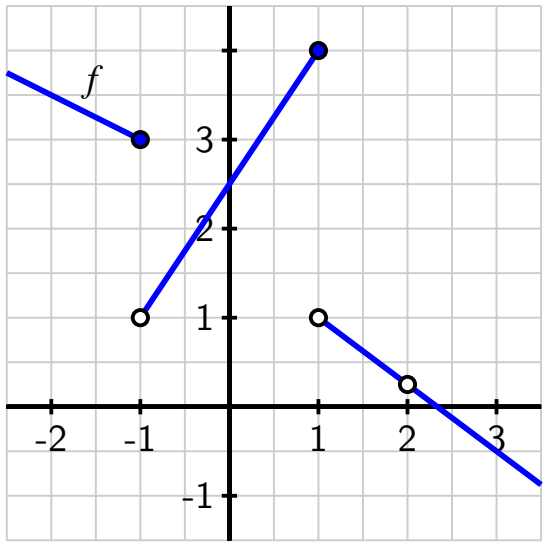
(c) Are there any values of x for which $r(x) = 0$? If not, explain why; if so, determine all such values, with justification.

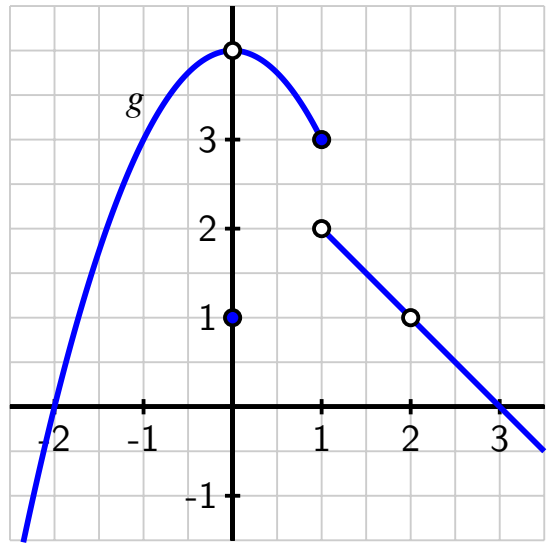
(d) Let $k(x) = f(x) \cdot g(x)$. Determine $k(0)$.

(e) Let $s(x) = \frac{p(x)}{q(x)}$. Determine $s(1)$ exactly.

(f) Are there any values of x in the interval $-4 \leq x \leq 4$ for which $s(x)$ is not defined? If not, explain why; if so, determine all such values, with justification.

Activity 1.9.2. Consider the functions f and g defined by the following figures. Assume that the given lines and curves pass through intersection points on the grid when it looks plausible. For instance, $(0, 2.5)$ and $(3, -0.5)$ lie on the graph of f, and $(-1, 3)$ and $(1.5, 1.5)$ lie on the graph of g.





(a) Determine the exact value of $(f + g)(0)$.

(b) Determine the exact value of $(g - f)(1)$.

(c) Determine the exact value of $(f \cdot g)(-1)$.

(d) Are there any values of x for which $\left(\frac{f}{g}\right)(x)$ is undefined? If not, explain why. If so, determine the values and justify your answer.

(e) For what values of x is $(f \cdot g)(x) = 0$? Why?

(f) Are there any values of x for which $(f - g)(x) = 0$? Why or why not?

Activity 1.9.3. Let f be a function that measures a car's fuel economy in the following way. Given an input velocity v in miles per hour, $f(v)$ is the number of gallons of fuel that the car consumes per mile (i.e., "gallons per mile"). We know that $f(60) = 0.04$.

(a) What is the meaning of the statement "$f(60) = 0.04$" in the context of the problem? That is, what does this say about the car's fuel economy? Write a complete sentence to explain.

(b) Consider the function $g(v) = \frac{1}{f(v)}$. What is the value of $g(60)$? What are the units on g? What does g measure?

(c) Consider the function $h(v) = v \cdot f(v)$. What is the value of $h(60)$? What are the units on h? What does h measure?

(d) Do $f(60)$, $g(60)$, and $h(60)$ tell us fundamentally different information, or are they all essentially saying the same thing? Explain.

(e) Suppose we also know that $f(70) = 0.045$. Find the average rate of change of f on the interval $[60, 70]$. What are the units on the average rate of change of f? What does this quantity measure? Write a complete sentence to explain.

Activity 1.9.4. In what follows, we work to understand two different piecewise functions entirely by hand based on familiar properties of linear and quadratic functions.

(a) Consider the function p defined by the following rule:

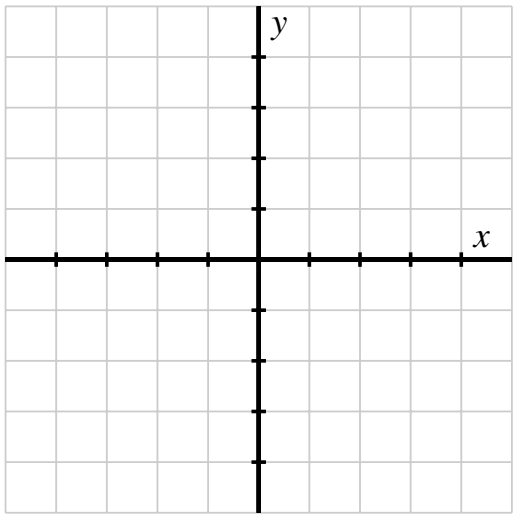
$$p(x) = \begin{cases} -(x+2)^2 + 2, & x < 0 \\ \frac{1}{2}(x-2)^2 + 1, & x \geq 0 \end{cases}$$

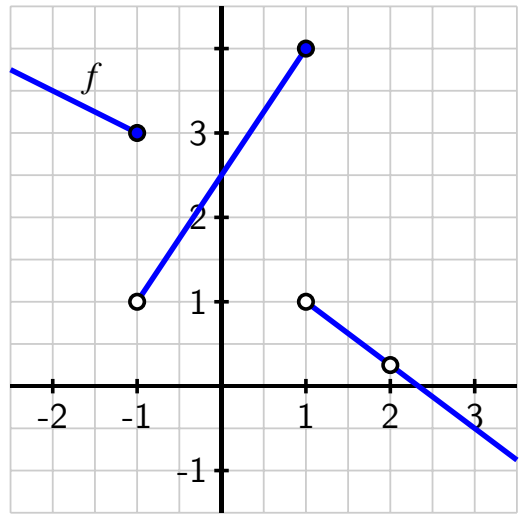
What are the values of $p(-4)$, $p(-2)$, $p(0)$, $p(2)$, and $p(4)$?

(b) What point is the vertex of the quadratic part of p that is valid for $x < 0$? What point is the vertex of the quadratic part of p that is valid for $x \geq 0$?

(c) For what values of x is $p(x) = 0$? In addition, what is the y-intercept of p?

(d) Sketch an accurate, labeled graph of $y = p(x)$ on the axes provided at left in the following figure.





(e) For the function f defined by the righthand figure in (d), determine a piecewise-defined formula for f that is expressed in bracket notation similar to the definition of $y = p(x)$ above.

CHAPTER 2

Circular Functions

2.1 Traversing Circles

Preview Activity 2.1.1. In the context of the ferris wheel pictured in Figure 2.1.1 in the text, assume that the height, h, of the moving point (the cab in which you are riding), and the distance, d, that the point has traveled around the circumference of the ferris wheel are both measured in meters.
Further, assume that the circumference of the ferris wheel is 150 meters. In addition, suppose that after getting in your cab at the lowest point on the wheel, you traverse the full circle several times.

(a) Recall that the circumference, C, of a circle is connected to the circle's radius, r, by the formula $C = 2\pi r$. What is the radius of the ferris wheel? How high is the highest point on the ferris wheel?

(b) How high is the cab after it has traveled $1/4$ of the circumference of the circle?

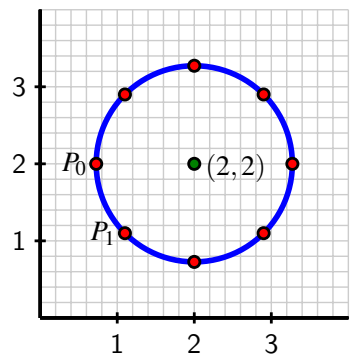
(c) How much distance along the circle has the cab traversed at the moment it first reaches a height of $\frac{150}{\pi} \approx 47.75$ meters?

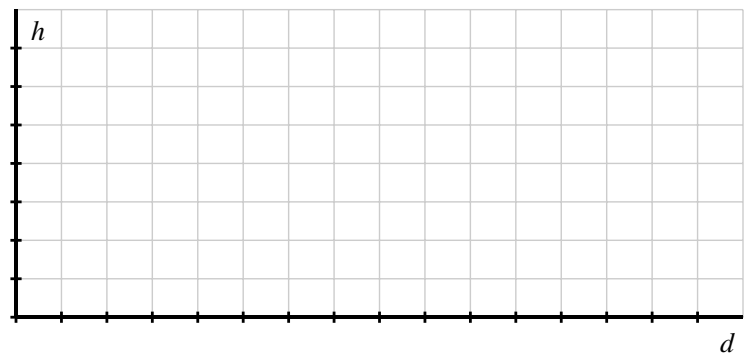
(d) Can h be thought of as a function of d? Why or why not?

(e) Can d be thought of as a function of h? Why or why not?

(f) Why do you think the curve shown at right in Figure 2.1.1 has the shape that it does? Write several sentences to explain.

Activity 2.1.2. Consider the circle pictured in the following figure that is centered at the point $(2, 2)$ and that has circumference 8. Assume that we track the y-coordinate (that is, the height, h) of a point that is traversing the circle counterclockwise and that it starts at P_0 as pictured.





(a) How far along the circle is the point P_1 from P_0? Why?

(b) Label the subsequent points in the figure P_2, P_3, ... as we move counterclockwise around the circle. What is the exact y-coordinate of the point P_2? of P_4? Why?

(c) Determine the y-coordinates of the remaining points on the circle (exactly where possible, otherwise approximately) and hence complete the entries in the following table that track the height, h, of the point traversing the circle as a function of distance traveled, d. Note that the d-values in the table correspond to the point traversing the circle more than once.

d	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
h	2																

(d) By plotting the points in the table in part (c) and connecting them in an intuitive way, sketch a graph of h as a function of d over the interval $0 \leq d \leq 16$ on the axes provided in the figure in part (a). Clearly label the scale of your axes and the coordinates of several important points on the curve.

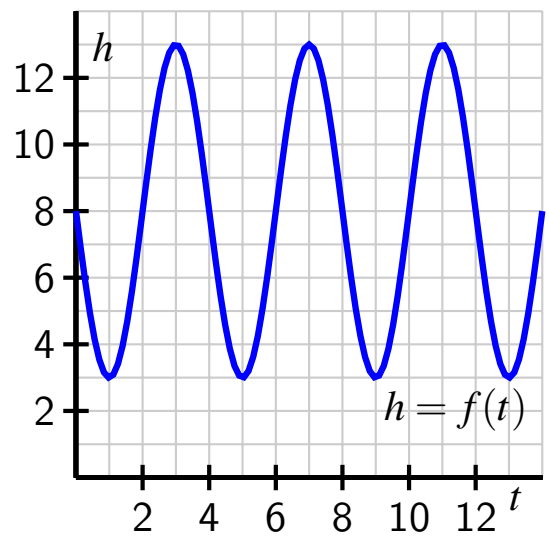
(e) What is similar about your graph in comparison to the one in Figure 2.1.5 in the text? What is different?

(f) What will be the value of h when $d = 51$? How about when $d = 102$?

Activity 2.1.3. A weight is placed on a frictionless table next to a wall and attached to a spring that is fixed to the wall. From its natural position of rest, the weight is imparted an initial velocity that sets it in motion. The weight then oscillates back and forth, and we can measure its distance, $h = f(t)$ (in inches) from the wall at any given time, t (in seconds). A graph of f and a table of select values are given below.

t	$f(t)$
0.25	6.087
0.5	4.464
0.75	3.381
1	3.000
1.25	3.381
1.5	4.464
1.75	6.087
2	8.000

t	$f(t)$
2.25	9.913
2.5	11.536
2.75	12.619
3	13.000
3.25	12.619
3.5	11.536
3.75	9.913
4	8.000



(a) Determine the period p, midline $y = m$, and amplitude a of the function f.

(b) What is the greatest distance the weight is displaced from the wall? What is the least distance the weight is displaced from the wall? What is the range of f?

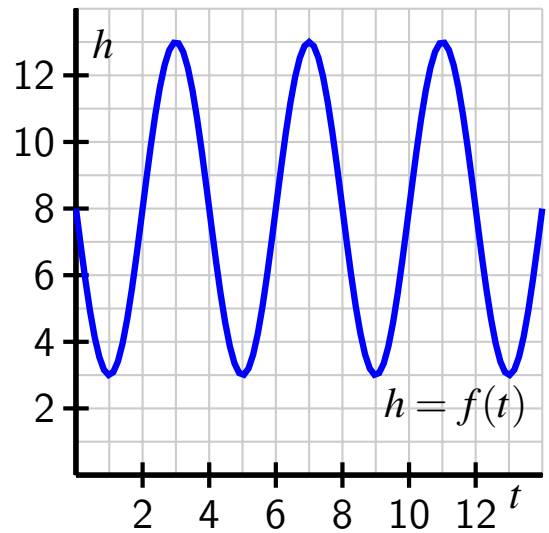
(c) Determine the average rate of change of f on the intervals $[4, 4.25]$ and $[4.75, 5]$. Write one careful sentence to explain the meaning of each (including units). In addition, write a sentence to compare the two different values you find and what they together say about the motion of the weight.

(d) Based on the periodicity of the function, what is the value of $f(6.75)$? of $f(11.25)$?

Activity 2.1.4. Consider the same setting as Activity 2.1.3: a weight oscillates back and forth on a frictionless table with distance from the wall given by, $h = f(t)$ (in inches) at any given time, t (in seconds). A graph of f and a table of select values are given below.

t	$f(t)$
0.25	6.087
0.5	4.464
0.75	3.381
1	3.000
1.25	3.381
1.5	4.464
1.75	6.087
2	8.000

t	$f(t)$
2.25	9.913
2.5	11.536
2.75	12.619
3	13.000
3.25	12.619
3.5	11.536
3.75	9.913
4	8.000



(a) Determine $AV_{[2,2.25]}$, $AV_{[2.25,2.5]}$, $AV_{[2.5,2.75]}$, and $AV_{[2.75,3]}$. What do these four values tell us about how the weight is moving on the interval $[2, 3]$?

(b) Give an example of an interval of length 0.25 units on which f has its most negative average rate of change. Justify your choice.

(c) Give an example of the longest interval you can find on which f is decreasing.

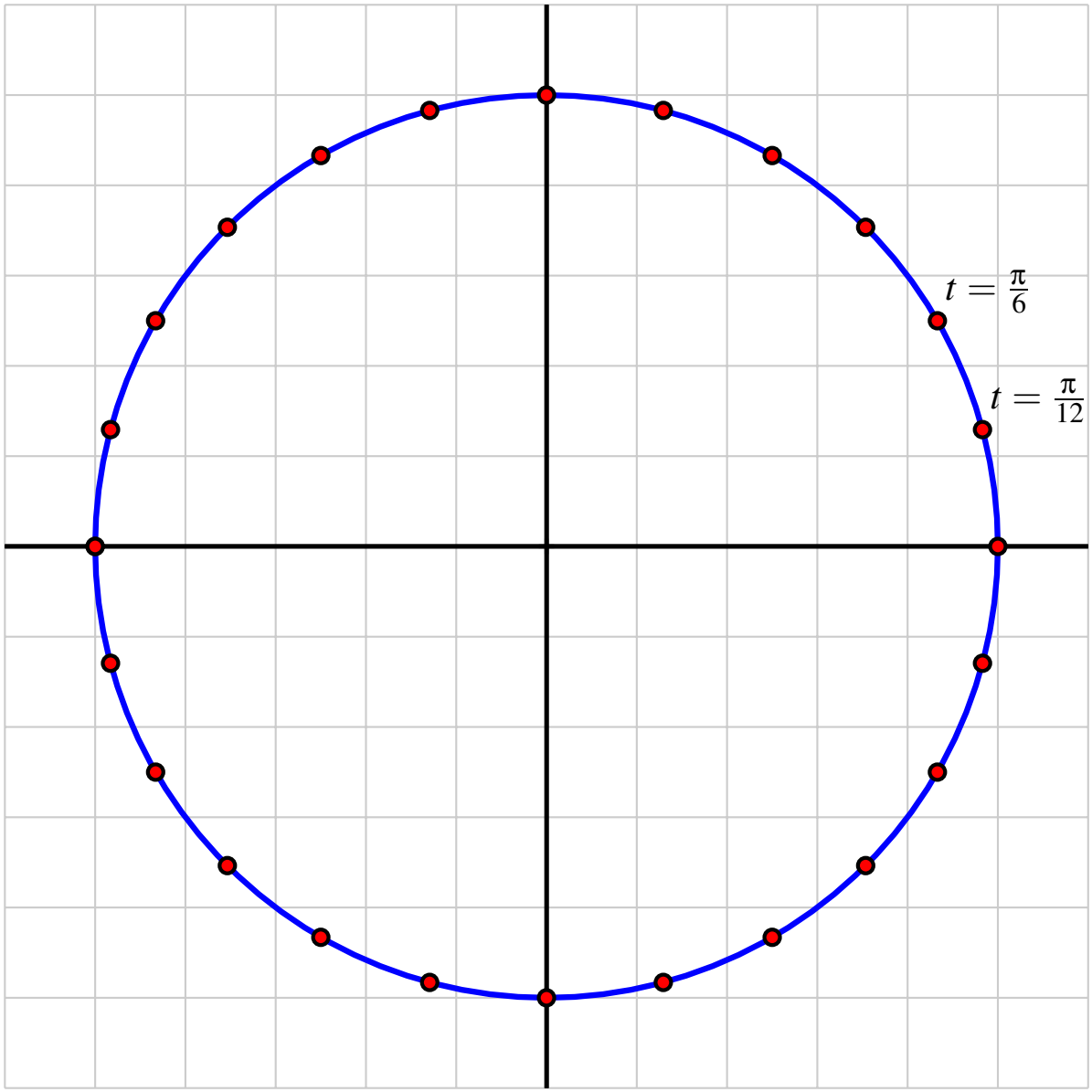
(d) Give an example of an interval on which f is concave up. (Recall that a function is concave up on an interval provided that throughout the interval, the curve bends upward, similar to a parabola that opens up.)

(e) On an interval where f is both decreasing and concave down, what does this tell us about how the weight is moving on that interval? For instance, is the weight moving toward or away from the wall? is it speeding up or slowing down?

(f) What general conclusions can you make about the average rate of change of a circular function on intervals near its highest or lowest points? about its average rate of change on intervals near the function's midline?

2.2 The Unit Circle

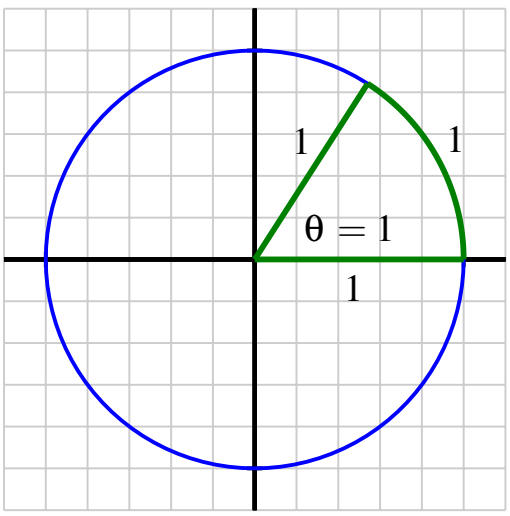
Preview Activity 2.2.1. In the following figure there are 24 equally spaced points on the unit circle. Since the circumference of the unit circle is 2π, each of the points is $\frac{1}{24} \cdot 2\pi = \frac{\pi}{12}$ units apart (traveled along the circle). Thus, the first point counterclockwise from $(1, 0)$ corresponds to the distance $t = \frac{\pi}{12}$ traveled along the unit circle. The second point is twice as far, and thus $t = 2 \cdot \frac{\pi}{12} = \frac{\pi}{6}$ units along the circle away from $(1, 0)$.



(a) Label each of the subsequent points on the unit circle with the exact distance they lie counter-clockwise away from $(1, 0)$; write each fraction in lowest terms.

(b) Which distance along the unit circle corresponds to $\frac{1}{4}$ of a full rotation around? to $\frac{5}{8}$ of a full rotation?

(c) One way to measure angles is connected to the arc length along a circle. For an angle whose vertex is at $(0, 0)$ in the unit circle, we say the angle's measure is 1 **radian** provided that the angle intercepts an arc of the circle that is 1 unit in length, as pictured in the following figure. Note particularly that an angle measuring 1 radian intercepts an arc of the same length as the circle's radius.



Suppose that α and β are angles with respective radian measures $\alpha = \frac{\pi}{3}$ and $\beta = \frac{3\pi}{4}$. Assuming that we view α and β as having their vertex at $(0, 0)$ and one side along the positive x-axis, sketch the angles α and β on the unit circle in part (a).

(d) What is the radian measure that corresponds to a $90°$ angle?

Activity 2.2.2. Convert each of the following quantities to the alternative measure: degrees to radians or radians to degrees.

(a) $30°$

(b) $\frac{2\pi}{3}$ radians

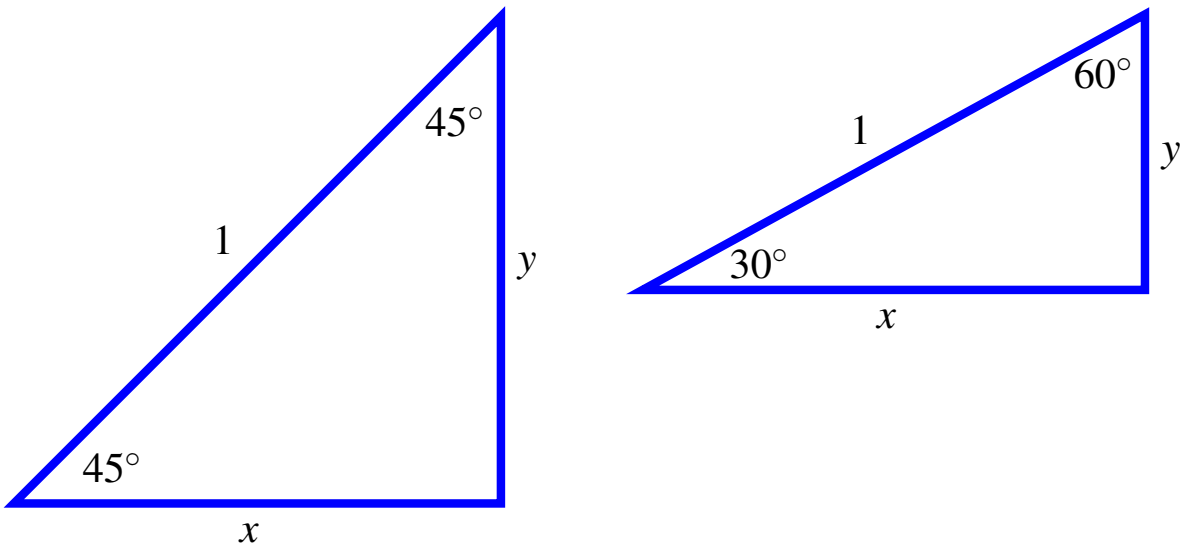
(c) $\frac{5\pi}{4}$ radians

(d) $240°$

(e) $17°$

(f) 2 radians

Activity 2.2.3. In what follows, we work to understand key relationships in $45°$-$45°$-$90°$ and $30°$-$60°$-$90°$ triangles.

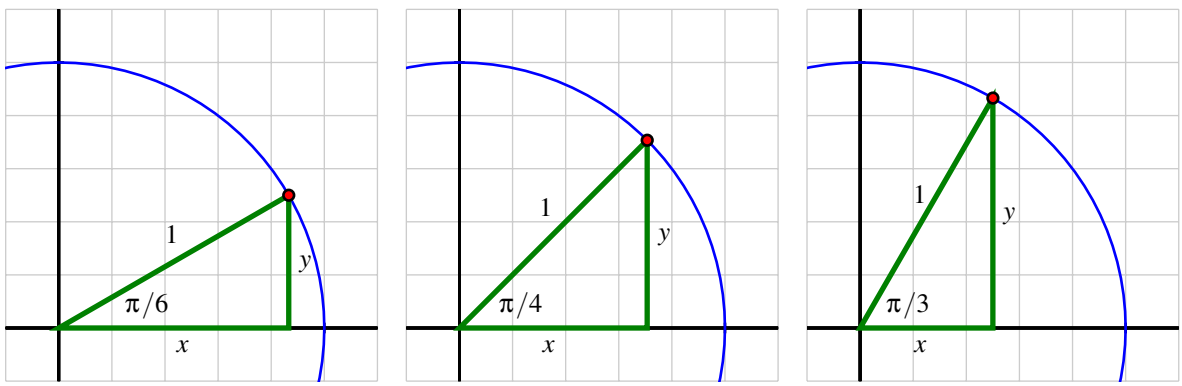


(a) For the $45°$-$45°$-$90°$ triangle with legs of length x and y and hypotenuse of length 1, what does the fact that the triangle is isosceles tell us about the relationship between x and y? What are their exact values?

(b) Now consider the $30°$-$60°$-$90°$ triangle with hypotenuse of length 1 and the longer leg (of length x) lying along the positive x-axis. What special kind of triangle is formed when we reflect this triangle across the x-axis? How can we use this perspective to determine the exact values of x and y?

(c) Suppose we consider the related $30°$-$60°$-$90°$ triangle with hypotenuse of length 1 and the shorter leg (of length x) lying along the positive x-axis. What are the *exact* values of x and y in this triangle?

(d) We know from the conversion factor from degrees to radians that an angle of $30°$ corresponds to an angle measuring $\frac{\pi}{6}$ radians, an angle of $45°$ corresponds to $\frac{\pi}{4}$ radians, and $60°$ corresponds to $\frac{\pi}{3}$ radians.



Use your work in (a), (b), and (c) to label the noted point in each of the three respective figures with its exact coordinates.

Activity 2.2.4. Determine each of the following values or points exactly.

(a) In a circle of radius 11, the arc length intercepted by a central angle of $\frac{5\pi}{3}$.

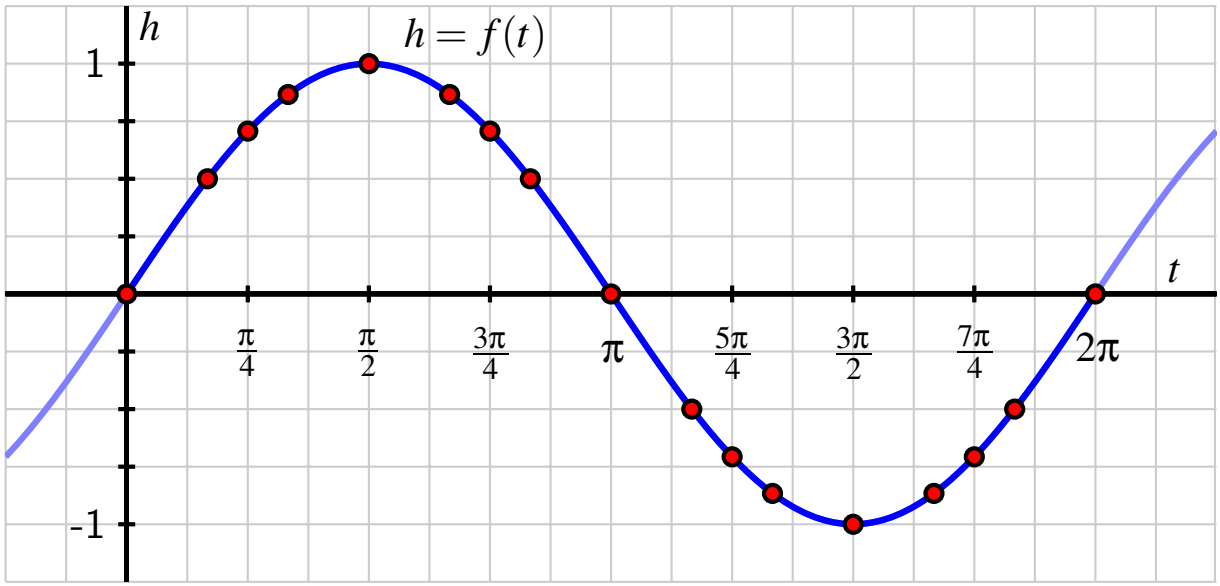
(b) In a circle of radius 3, the central angle measure that intercepts an arc of length $\frac{\pi}{4}$.

(c) The radius of the circle in which an angle of $\frac{7\pi}{6}$ intercepts an arc of length $\frac{\pi}{2}$.

(d) The exact coordinates of the point on the circle of radius 5 that lies $\frac{25\pi}{6}$ units counterclockwise along the circle from $(5, 0)$.

2.3 The Sine and Cosine Functions

Preview Activity 2.3.1. If we consider the unit circle with 16 labeled special points in Figure 2.3.1, start at $t = 0$, and traverse the circle counterclockwise, we may view the height, h, of the traversing point as a function of the angle, t, in radians. From there, we can plot the resulting (t, h) ordered pairs and connect them to generate the circular function pictured in the following figure, which tracks the height of a point traversing the unit circle.



(a) What is the exact value of $f(\frac{\pi}{4})$? of $f(\frac{\pi}{3})$?

(b) Complete the following table with the exact values of h that correspond to the stated inputs.

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
h									

t	π	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
h									

(c) What is the exact value of $f(\frac{11\pi}{4})$? of $f(\frac{14\pi}{3})$?

(d) Give four different values of t for which $f(t) = -\frac{\sqrt{3}}{2}$.

Activity 2.3.2. Let $k = g(t)$ be the function that tracks the x-coordinate of a point traversing the unit circle counterclockwise from $(1, 0)$. That is, $g(t) = \cos(t)$. Use the information we know about the unit circle that is summarized in Figure 2.3.1 (with 16 labeled special points) to respond to the following questions.

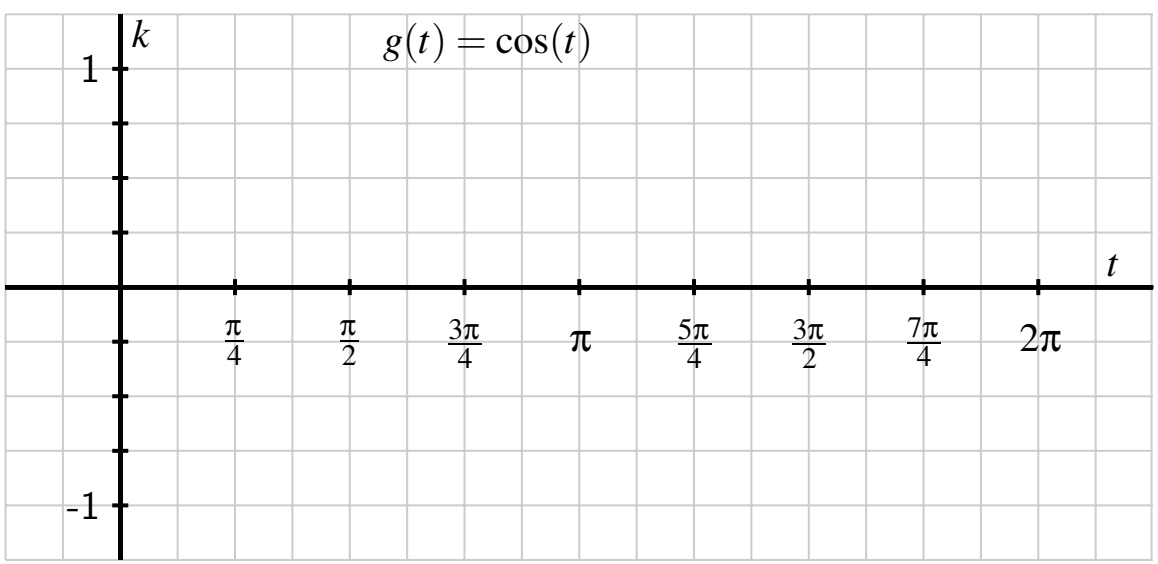
(a) What is the exact value of $\cos(\frac{\pi}{6})$? of $\cos(\frac{5\pi}{6})$? $\cos(-\frac{\pi}{3})$?

(b) Complete the following table with the exact values of k that correspond to the stated inputs.

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
k									

t	π	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
k									

(c) On the axes provided in the following figure, sketch an accurate graph of $k = \cos(t)$. Label the exact location of several key points on the curve.



(d) What is the exact value of $\cos(\frac{11\pi}{4})$? of $\cos(\frac{14\pi}{3})$?

(e) Give four different values of t for which $\cos(t) = -\frac{\sqrt{3}}{2}$.

(f) How is the graph of $k = \cos(t)$ different from the graph of $h = \sin(t)$? How are the graphs similar?

Activity 2.3.3. Use Figure 2.3.12 in the text (which plots the sine and cosine functions ont the same axes) to assist in answering the following questions.

(a) Give an example of the largest interval you can find on which $f(t) = \sin(t)$ is decreasing.

(b) Give an example of the largest interval you can find on which $f(t) = \sin(t)$ is decreasing and concave down.

(c) Give an example of the largest interval you can find on which $g(t) = \cos(t)$ is increasing.

(d) Give an example of the largest interval you can find on which $g(t) = \cos(t)$ is increasing and concave up.

(e) Without doing any computation, on which interval is the average rate of change of $g(t) = \cos(t)$ greater: $[\pi, \pi + 0.1]$ or $[\frac{3\pi}{2}, \frac{3\pi}{2} + 0.1]$? Why?

(f) In general, how would you characterize the locations on the sine and cosine graphs where the functions are increasing or decreasingly most rapidly?

(g) Thinking from the perspective of the unit circle, for which quadrants of the x-y plane is $\cos(t)$ negative for an angle t that lies in that quadrant?

Activity 2.3.4. Answer the following questions exactly wherever possible. If you estimate a value, do so to at least 5 decimal places of accuracy.

(a) The x coordinate of the point on the unit circle that lies in the third quadrant and whose y-coordinate is $y = -\frac{3}{4}$.

(b) The y-coordinate of the point on the unit circle generated by a central angle opening counterclockwise with one side on the positive x-axis that measures $t = 2$ radians.

(c) The x-coordinate of the point on the unit circle generated by a central angle with one side on the positive x-axis that measures $t = -3.05$ radians. (With the negative radian measure, we view the angle as opening counterclockwise from its initial side on the positive x-axis.)

(d) The value of $\cos(t)$ where t is an angle in Quadrant II that satisfies $\sin(t) = \frac{1}{2}$.

(e) The value of $\sin(t)$ where t is an angle in Quadrant III for which $\cos(t) = -0.7$.

(f) The average rate of change of $f(t) = \sin(t)$ on the intervals $[0.1, 0.2]$ and $[0.8, 0.9]$.

(g) The average rate of change of $g(t) = \cos(t)$ on the intervals $[0.1, 0.2]$ and $[0.8, 0.9]$.

2.4 Sinusoidal Functions

Preview Activity 2.4.1. Let $f(t) = \cos(t)$. First, answer all of the questions below *without* using *Desmos*; then use *Desmos* to confirm your conjectures. For each prompt, describe the graphs of g and h as transformations of f and, in addition, state the amplitude, midline, and period of both g and h.

(a) $g(t) = 3\cos(t)$ and $h(t) = -\frac{1}{4}\cos(t)$

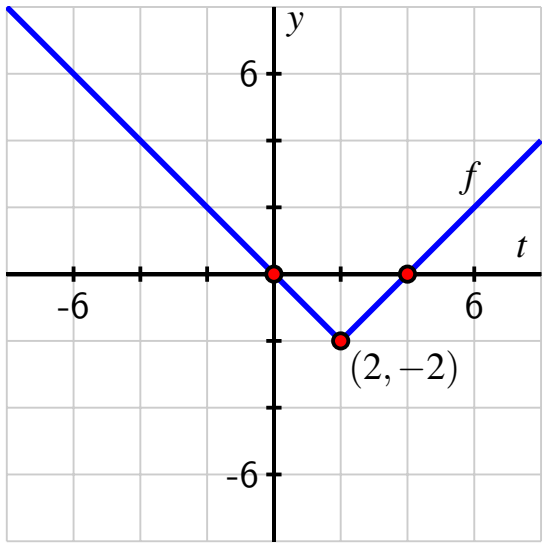
(b) $g(t) = \cos(t - \pi)$ and $h(t) = \cos\left(t + \frac{\pi}{2}\right)$

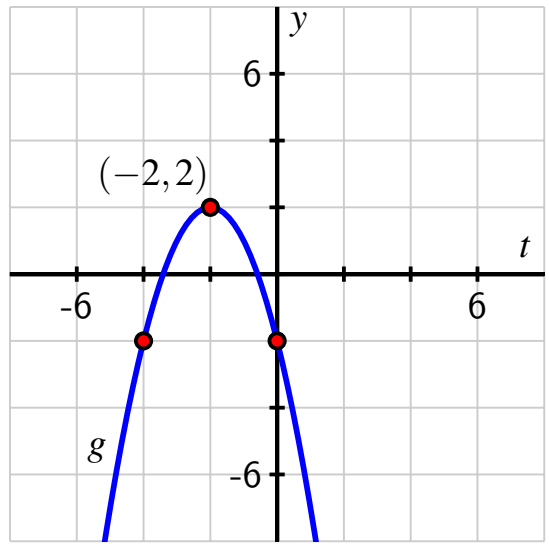
(c) $g(t) = \cos(t) + 4$ and $h(t) = \cos(t) - 2$

(d) $g(t) = 3\cos(t - \pi) + 4$ and $h(t) = -\frac{1}{4}\cos\left(t + \frac{\pi}{2}\right) - 2$

Activity 2.4.2. Consider a spring-mass system where a weight is resting on a frictionless table. We let $d(t)$ denote the distance from the wall (where the spring is attached) to the weight at time t in seconds and know that the weight oscillates periodically with a minimum value of 2 feet and a maximum value of 7 feet with a period of 2π. We also know that $d(0) = 4.5$ and $d\left(\frac{\pi}{2}\right) = 2$.
Determine a formula for $d(t)$ in the form $d(t) = a\cos(t - b) + c$ or $d(t) = a\sin(t - b) + c$. Is it possible to find two different formulas that work? For any formula you find, identify the anchor point.

Activity 2.4.3. Consider the functions f and g given in the following figures.

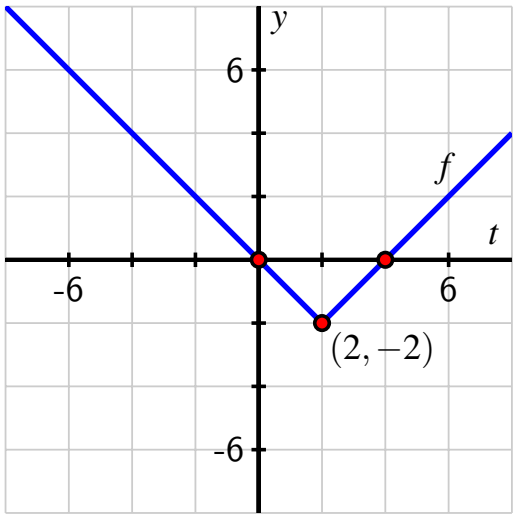


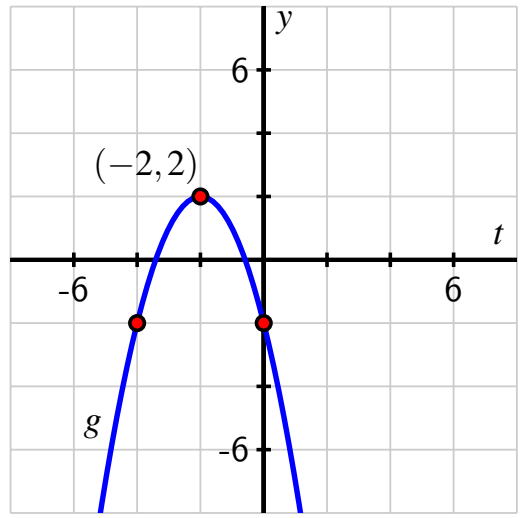


(a) On the same axes as the plot of $y = f(t)$, sketch the following graphs: $y = h(t) = f(\frac{1}{3}t)$ and $y = j(t) = f(4t)$. Be sure to label several points on each of f, h, and j with arrows to indicate their correspondence. In addition, write one sentence to explain the overall transformations that have resulted in h and j from f.

(b) On the same axes as the plot of $y = g(t)$, sketch the following graphs: $y = k(t) = g(2t)$ and $y = m(t) = g(\frac{1}{2}t)$. Be sure to label several points on each of g, k, and m with arrows to indicate their correspondence. In addition, write one sentence to explain the overall transformations that have resulted in k and m from g.

(c) On the additional copies of the two figures below, sketch the graphs of the following transformed functions: $y = r(t) = 2f(\frac{1}{2}t)$ (at left) and $y = s(t) = \frac{1}{2}g(2t)$. As above, be sure to label several points on each graph and indicate their correspondence to points on the original parent function.





(d) Describe in words how the function $y = r(t) = 2f(\frac{1}{2}t)$ is the result of composing two elementary transformations of $y = f(t)$. Does the order in which these transformations are composed matter? Why or why not?

Activity 2.4.4. Determine the exact period, amplitude, and midline of each of the following functions. In addition, state the range of each function, any horizontal shift that has been introduced to the graph, and identify an anchor point. Make your conclusions without consulting *Desmos*, and then use the program to check your work.

(a) $p(x) = \sin(10x) + 2$

(b) $q(x) = -3\cos(0.25x) - 4$

(c) $r(x) = 2\sin\left(\frac{\pi}{4}x\right) + 5$

(d) $w(x) = 2\cos\left(\frac{\pi}{2}(x-3)\right) + 5$

(e) $u(x) = -0.25\sin(3x-6) + 5$

Activity 2.4.5. Consider a spring-mass system where the weight is hanging from the ceiling in such a way that the following is known: we let $d(t)$ denote the distance from the ceiling to the weight at time t in seconds and know that the weight oscillates periodically with a minimum value of 1.5 feet and a maximum value of 4 feet, with a period of 3, and you know $d(0.5) = 2.75$ and $d(1.25) = 4$.
State the midline, amplitude, range, and an anchor point for the function, and hence determine a formula for $d(t)$ in the form $a\cos(k(t-b)) + c$ or $a\sin(k(t-b)) + c$. Show your work and thinking, and use *Desmos* appropriately to check that your formula generates the desired behavior.

CHAPTER 3

Exponential and Logarithmic Functions

3.1 Exponential Growth and Decay

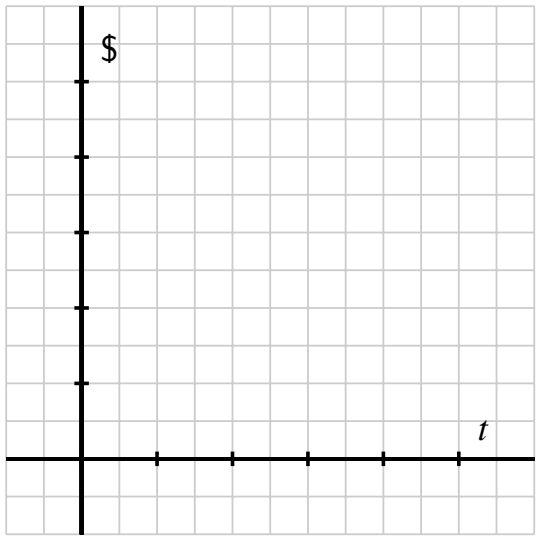
Preview Activity 3.1.1. Suppose that at age 20 you have \$20000 and you can choose between one of two ways to use the money: you can invest it in a mutual fund that will, on average, earn 8% interest annually, or you can purchase a new automobile that will, on average, depreciate 12% annually. Let's explore how the \$20000 changes over time.
Let $I(t)$ denote the value of the \$20000 after t years if it is invested in the mutual fund, and let $V(t)$ denote the value of the automobile t years after it is purchased.

(a) Determine $I(0)$, $I(1)$, $I(2)$, and $I(3)$.

(b) Note that if a quantity depreciates 12% annually, after a given year, 88% of the quantity remains. Compute $V(0)$, $V(1)$, $V(2)$, and $V(3)$.

(c) Based on the patterns in your computations in (a) and (b), determine formulas for $I(t)$ and $V(t)$.

(d) Use *Desmos* to define $I(t)$ and $V(t)$. Plot each function on the interval $0 \leq t \leq 20$ and record your results on the axes inthe figure below, being sure to label the scale on the axes. What trends do you observe in the graphs? How do $I(20)$ and $V(20)$ compare?



Activity 3.1.2. In *Desmos*, define the function $g(t) = ab^t$ and create sliders for both a and b when prompted. Click on the sliders to set the minimum value for each to 0.1 and the maximum value to 10. Note that for g to be an exponential function, we require $b \neq 1$, even though the slider for b will allow this value.

(a) What is the domain of $g(t) = ab^t$?

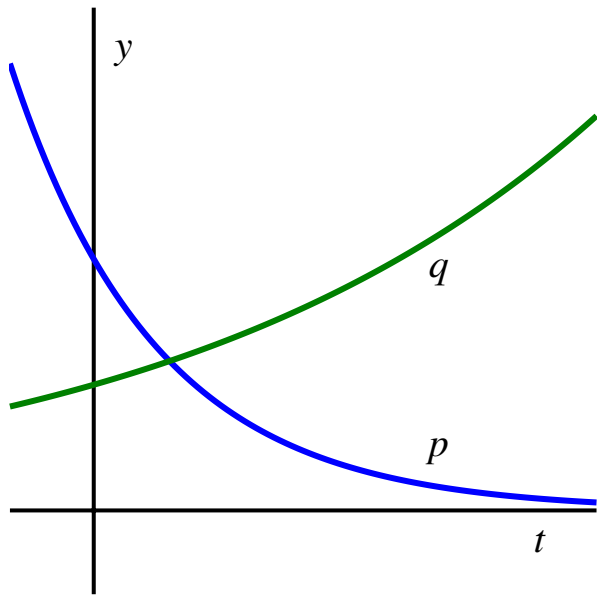
(b) What is the range of $g(t) = ab^t$?

(c) What is the y-intercept of $g(t) = ab^t$?

(d) How does changing the value of b affect the shape and behavior of the graph of $g(t) = ab^t$? Write several sentences to explain.

(e) For what values of the growth factor b is the corresponding growth rate positive? For which b-values is the growth rate negative?

(f) Consider the graphs of the exponential functions p and q provided in the following figure. If $p(t) = ab^t$ and $q(t) = cd^t$, what can you say about the values a, b, c, and d (beyond the fact that all are positive and $b \neq 1$ and $d \neq 1$)? For instance, can you say a certain value is larger than another? Or that one of the values is less than 1?



Activity 3.1.3. The value of an automobile is depreciating. When the car is 3 years old, its value is \$12500; when the car is 7 years old, its value is \$6500.

(a) Suppose the car's value t years after its purchase is given by the function $V(t)$ and that V is exponential with form $V(t) = ab^t$, what are the values of a and b? Find a and b both exactly and approximately.

(b) Using the exponential model determined in (a), determine the purchase value of the car and then use *Desmos* to estimate when the car will be worth less than \$1000.

(c) Suppose instead that the car's value is modeled by a linear function L and satisfies the values stated at the outset of this activity. Find a formula for $L(t)$ and determine both the purchase value of the car and when the car will be worth \$1000.

(d) Which model do you think is more realistic? Why?

Activity 3.1.4. For each of the following prompts, give an example of a function that satisfies the stated characteristics by both providing a formula and sketching a graph.

(a) A function p that is always decreasing and decreases at a constant rate.

(b) A function q that is always increasing and increases at an increasing rate.

(c) A function r that is always increasing for $t < 2$, always decreasing for $t > 2$, and is always changing at a decreasing rate.

(d) A function s that is always increasing and increases at a decreasing rate. (Hint: to find a formula, think about how you might use a transformation of a familiar function.)

(e) A function u that is always decreasing and decreases at a decreasing rate.

3.2 Modeling with exponential functions

Preview Activity 3.2.1. In *Desmos*, define $g(t) = ab^t + c$ and accept the prompt for sliders for a, b, and c. Edit the sliders so that a has values from $a = 5$ to $a = 50$, b has values from $b = 0.7$ to $b = 1.3$, and c has values from $c = -5$ to $c = 5$ (each with a step-size of 0.01). In addition, in *Desmos* let $P = (0, g(0))$ and check the box to show the label. Finally, zoom out so that the window shows an interval of t-values from $-30 \leq t \leq 30$.

(a) Set $b = 1.1$ and explore the effects of changing the values of a and c. Write several sentences to summarize your observations.

(b) Follow the directions for (a) again, this time with $b = 0.9$

(c) Set $a = 5$ and $c = 4$. Explore the effects of changing the value of b; be sure to include values of b both less than and greater than 1. Write several sentences to summarize your observations.

(d) When $0 < b < 1$, what happens to the graph of g when we consider positive t-values that get larger and larger?

Activity 3.2.2. For each of the following functions, *without* using graphing technology, determine whether the function is

i. always increasing or always decreasing;

ii. always concave up or always concave down; and

iii. increasing without bound, decreasing without bound, or increasing/decreasing toward a finite value.

In addition, state the y-intercept and the range of the function. For each function, write a sentence that explains your thinking and sketch a rough graph of how the function appears.

(a) $p(t) = 4372(1.000235)^t + 92856$

(b) $q(t) = 27931(0.97231)^t + 549786$

(c) $r(t) = -17398(0.85234)^t$

(d) $s(t) = -17398(0.85234)^t + 19411$

(e) $u(t) = -7522(1.03817)^t$

(f) $v(t) = -7522(1.03817)^t + 6731$

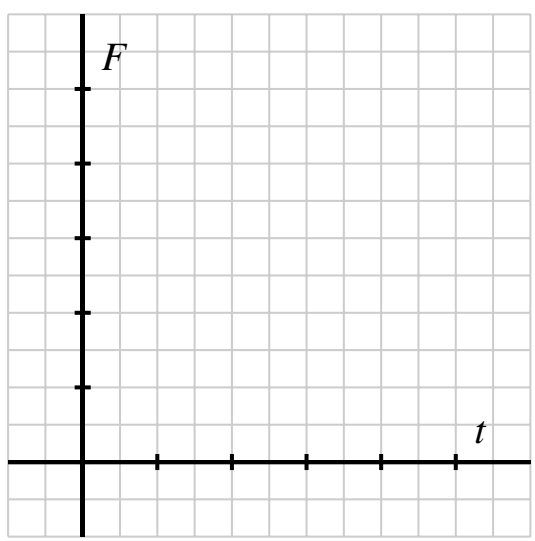
Activity 3.2.3. A can of soda (at room temperature) is placed in a refrigerator at time $t = 0$ (in minutes) and its temperature, $F(t)$, in degrees Fahrenheit, is computed at regular intervals. Based on the data, a model is formulated for the object's temperature, given by

$$F(t) = 42 + 30(0.95)^t.$$

(a) Consider the simpler (parent) function $p(t) = (0.95)^t$. How do you expect the graph of this function to appear? How will it behave as time increases? Without using graphing technology, sketch a rough graph of p and write a sentence of explanation.

(b) For the slightly more complicated function $r(t) = 30(0.95)^t$, how do you expect this function to look in comparison to p? What is the long-range behavior of this function as t increases? Without using graphing technology, sketch a rough graph of r and write a sentence of explanation.

(c) Finally, how do you expect the graph of $F(t) = 42+30(0.95)^t$ to appear? Why? First sketch a rough graph without graphing technology, and then use technology to check your thinking and report an accurate, labeled graph on the axes provided below.



(d) What is the temperature of the refrigerator? What is the room temperature of the surroundings outside the refrigerator? Why?

(e) Determine the average rate of change of F on the intervals $[10, 20]$, $[20, 30]$, and $[30, 40]$. Write at least two careful sentences that explain the meaning of the values you found, including units, and discuss any overall trend in how the average rate of change is changing.

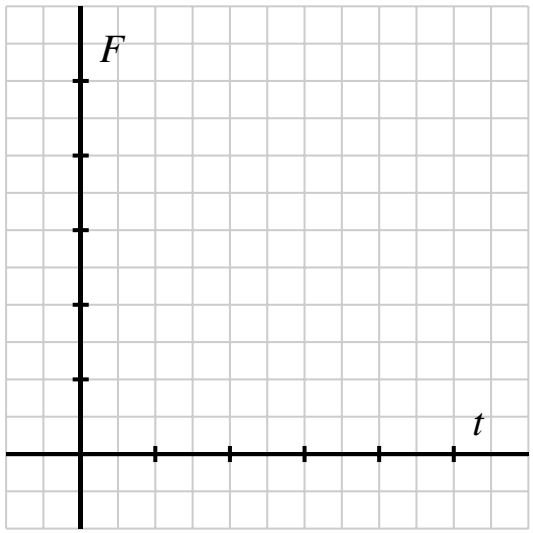
Activity 3.2.4. A potato initially at room temperature ($68°$) is placed in an oven (at $350°$) at time $t = 0$. It is known that the potato's temperature at time t is given by the function $F(t) = a - b(0.98)^t$ for some positive constants a and b, where F is measured in degrees Fahrenheit and t is time in minutes.

(a) What is the numerical value of $F(0)$? What does this tell you about the value of $a - b$?

(b) Based on the context of the problem, what should be the long-range behavior of the function $F(t)$? Use this fact along with the behavior of $(0.98)^t$ to determine the value of a. Write a sentence to explain your thinking.

(c) What is the value of b? Why?

(d) Check your work above by plotting the function F using graphing technology in an appropriate window. Record your results on the axes provided below, labeling the scale on the axes. Then, use the graph to estimate the time at which the potato's temperature reaches 325 degrees.



(e) How can we view the function $F(t) = a - b(0.98)^t$ as a transformation of the parent function $f(t) = (0.98)^t$? Explain.

3.3 The special number e

Preview Activity 3.3.1. Open a new *Desmos* worksheet and define the following functions: $f(t) = 2^t$, $g(t) = 3^t$, $h(t) = (\frac{1}{3})^t$, and $p(t) = f(kt)$. After you define p, accept the slider for k, and set the range of the slider to be $-2 \leq k \leq 2$.

(a) By experimenting with the value of k, find a value of k so that the graph of $p(t) = f(kt) = 2^{kt}$ appears to align with the graph of $g(t) = 3^t$. What is the value of k?

(b) Similarly, experiment to find a value of k so that the graph of $p(t) = f(kt) = 2^{kt}$ appears to align with the graph of $h(t) = (\frac{1}{3})^t$. What is the value of k?

(c) For the value of k you determined in (a), compute 2^k. What do you observe?

(d) For the value of k you determined in (b), compute 2^k. What do you observe?

(e) Given any exponential function of the form b^t, do you think it's possible to find a value of k to that $p(t) = f(kt) = 2^{kt}$ is the same function as b^t? Why or why not?

Activity 3.3.2. Recall from Section 1.3 that the average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}.$$

In Section 1.6, we also saw that if we instead think of the average rate of change of f on the interval $[a, a + h]$, the expression changes to

$$AV_{[a,a+h]} = \frac{f(a + h) - f(a)}{h}.$$

In this activity we explore the average rate of change of $f(t) = e^t$ near the points where $t = 1$ and $t = 2$. In a new *Desmos* worksheet, let $f(t) = e^t$ and define the function A by the rule

$$A(h) = \frac{f(1 + h) - f(1)}{h}.$$

(a) What is the meaning of $A(0.5)$ in terms of the function f and its graph?

(b) Compute the value of $A(h)$ for at least 6 different small values of h, both positive and negative. For instance, one value to try might be $h = 0.0001$. Record a table of your results.

(c) What do you notice about the values you found in (b)? How do they compare to an important number?

(d) Explain why the following sentence makes sense: "The function e^t is increasing at an average rate that is about the same as its value on small intervals near $t = 1$."

(e) Adjust your definition of A in *Desmos* by changing 1 to 2 so that

$$A(h) = \frac{f(2+h) - f(2)}{h}.$$

How does the value of $A(h)$ compare to $f(2)$ for small values of h?

Activity 3.3.3. By graphing $f(t) = e^t$ and appropriate horizontal lines, estimate the solution to each of the following equations. Note that in some parts, you may need to do some algebraic work in addition to using the graph.

(a) $e^t = 2$

(b) $e^{3t} = 5$

(c) $2e^t - 4 = 7$

(d) $3e^{0.25t} + 2 = 6$

(e) $4 - 2e^{-0.7t} = 3$

(f) $2e^{1.2t} = 1.5e^{1.6t}$

3.4 What a logarithm is

Preview Activity 3.4.1. Let $P(t)$ be the "powers of 10" function, which is given by $P(t) = 10^t$.

(a) Complete the followion table to generate certain values of P.

t	-3	-2	-1	0	1	2	3
$y = P(t) = 10^t$							

(b) Why does P have an inverse function?

(c) Since P has an inverse function, we know there exists some other function, say L, such that writing "$y = P(t)$" says the exact same thing as writing "$t = L(y)$". In words, where P produces the result of raising 10 to a given power, the function L reverses this process and instead tells us the power to which we need to raise 10, given a desired result. Complete the followin table to generate a collection of values of L.

y	10^{-3}	10^{-2}	10^{-1}	10^0	10^1	10^2	10^3
$L(y)$							

(d) What are the domain and range of the function P? What are the domain and range of the function L?

Activity 3.4.2. For each of the following equations, determine the exact value of the unknown variable. If the exact value involves a logarithm, use a computational device to also report an approximate value. For instance, if the exact value is $y = \log_{10}(2)$, you can also note that $y \approx 0.301$.

(a) $10^t = 0.00001$

(b) $\log_{10}(1000000) = t$

(c) $10^t = 37$

(d) $\log_{10}(y) = 1.375$

(e) $10^t = 0.04$

(f) $3 \cdot 10^t + 11 = 147$

(g) $2\log_{10}(y) + 5 = 1$

Activity 3.4.3. Let $E(t) = e^t$ and $N(y) = \ln(y)$ be the natural exponential function and the natural logarithm function, respectively.

(a) What are the domain and range of E?

(b) What are the domain and range of N?

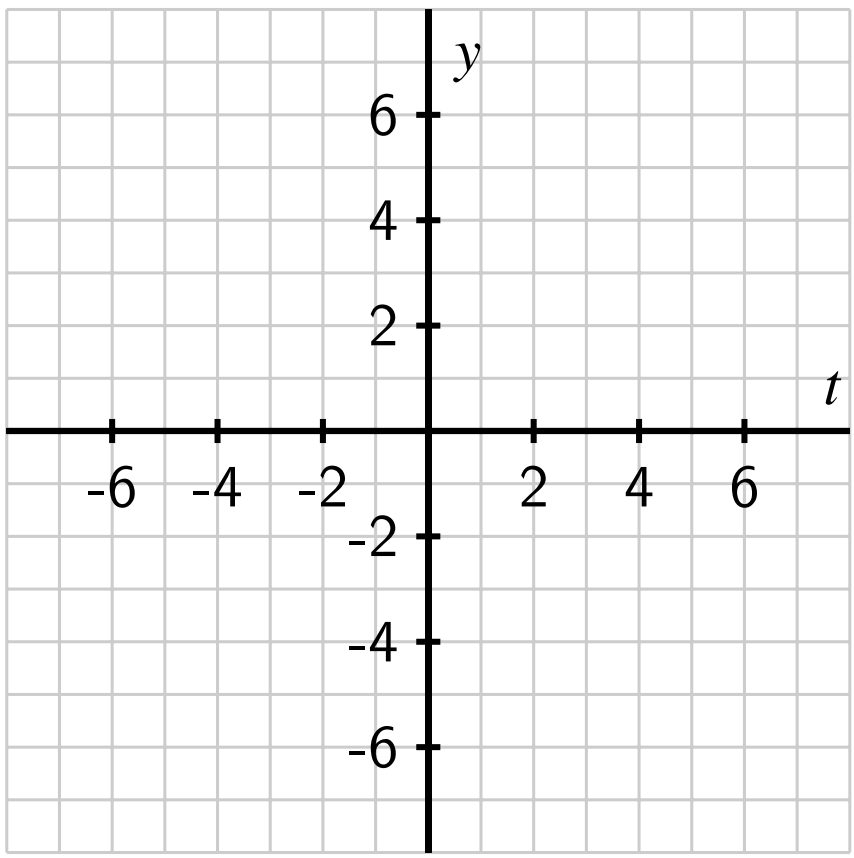
(c) What can you say about $\ln(e^t)$ for every real number t?

(d) What can you say about $e^{\ln(y)}$ for every positive real number y?

(e) Complete the following tables with both exact and approximate values of E and N. Then, plot the corresponding ordered pairs from each table on the axes provided and connect the points in an intuitive way. When you plot the ordered pairs on the axes, in both cases view the first line of the table as generating values on the horizontal axis and the second line of the table as producing values on the vertical axis. Note that when we take this perspective for plotting the data in the table for N, we are viewing N as a function of t, writing $N(t) = \ln(t)$ in order to plot the function on the t-y axes; label each ordered pair you plot appropriately.

t	-2	-1	0	1	2
$E(t) = e^t$	$e^{-2} \approx 0.135$				

y	e^{-2}	e^{-1}	1	e^1	e^2
$N(y) = \ln(y)$	-2				



Activity 3.4.4. Solve each of the following equations for the exact value of the unknown variable. If there is no solution to the equation, explain why not.

(a) $e^t = \frac{1}{10}$

(b) $5e^t = 7$

(c) $\ln(t) = -\frac{1}{3}$

(d) $e^{1-3t} = 4$

(e) $2\ln(t) + 1 = 4$

(f) $4 - 3e^{2t} = 2$

(g) $4 + 3e^{2t} = 2$

(h) $\ln(5 - 6t) = -2$

3.5 Properties and applications of logarithmic functions

Preview Activity 3.5.1. In the following questions, we investigate how $\log_{10}(a \cdot b)$ can be equivalently written in terms of $\log_{10}(a)$ and $\log_{10}(b)$.

(a) Write $10^x \cdot 10^y$ as 10 raised to a single power. That is, complete the equation

$$10^x \cdot 10^y = 10^{\square}$$

by filling in the box with an appropriate expression involving x and y.

(b) What is the simplest possible way to write $\log_{10} 10^x$? What about the simplest equivalent expression for $\log_{10} 10^y$?

(c) Explain why each of the following three equal signs is valid in the sequence of equalities:

$$\begin{aligned}\log_{10}(10^x \cdot 10^y) &= \log_{10}(10^{x+y}) \\ &= x + y \\ &= \log_{10}(10^x) + \log_{10}(10^y).\end{aligned}$$

(d) Suppose that a and b are positive real numbers so we can think of a as 10^x for some real number x and b as 10^y for some real number y. That is, say that $a = 10^x$ and $b = 10^y$. What does our work in (c) tell us about $\log_{10}(ab)$?

Activity 3.5.2. Solve each of the following equations exactly and then find an estimate that is accurate to 5 decimal places.

(a) $3^t = 5$

(b) $4 \cdot 2^t - 2 = 3$

(c) $3.7 \cdot (0.9)^{0.3t} + 1.5 = 2.1$

(d) $72 - 30(0.7)^{0.05t} = 60$

(e) $\ln(t) = -2$

(f) $3 + 2\log_{10}(t) = 3.5$

Activity 3.5.3. In the questions that follow, we compare and contrast the properties and behaviors of exponential and logarithmic functions.

(a) Let $f(t) = 1 - e^{-(t-1)}$ and $g(t) = \ln(t)$. Plot each function on the same set of coordinate axes. What properties do the two functions have in common? For what properties do the two functions differ? Consider each function's domain, range, t-intercept, y-intercept, increasing/decreasing behavior, concavity, and long-term behavior.

(b) Let $h(t) = a - be^{-k(t-c)}$, where a, b, c, and k are positive constants. Describe h as a transformation of the function $E(t) = e^t$.

(c) Let $r(t) = a + b\ln(t - c)$, where a, b, and c are positive constants. Describe r as a transformation of the function $L(t) = \ln(t)$.

(d) Data for the height of a tree is given in the following table; time t is measured in years and height is given in feet. At http://gvsu.edu/s/0yy (`gvsu.edu/s/0yy`), you can find a *Desmos* worksheet with this data already input.

t	1	2	3	4	5	6	7	8	9	10	11
$h(t)$	6	9.5	13	15	16.5	17.5	18.5	19	19.5	19.7	19.8

Do you think this data is better modeled by a logarithmic function of form $p(t) = a + b\ln(t - c)$ or by an exponential function of form $q(t) = m + ne^{-rt}$. Provide reasons based in how the data appears and how you think a tree grows, as well as by experimenting with sliders appropriately in *Desmos*. (Note: you may need to adjust the upper and lower bounds of several of the sliders in order to match the data well.)

Activity 3.5.4. Solve each of the following equations for the exact value of k.

(a) $41 = 50e^{-k \cdot 7}$

(b) $65 = 34 + 47e^{-k \cdot 45}$

(c) $7e^{2k-1} + 4 = 32$

(d) $\frac{5}{1+2e^{-10k}} = 4$

3.6 Modeling temperature and population

Preview Activity 3.6.1. In each of the following situations, determine the exact value of the unknown quantity that is identified.

(a) The temperature of a warming object in an oven is given by $F(t) = 275 - 203e^{-kt}$, and we know that the object's temperature after 20 minutes is $F(20) = 101$. Determine the exact value of k.

(b) The temperature of a cooling object in a refrigerator is modeled by $F(t) = a + 37.4e^{-0.05t}$, and the temperature of the refrigerator is $39.8°$. By thinking about the long-term behavior of $e^{-0.05t}$ and the long-term behavior of the object's temperature, determine the exact value of a.

(c) Later in this section, we'll learn that one model for how a population grows over time can be given by a function of the form

$$P(t) = \frac{A}{1 + Me^{-kt}}.$$

Models of this form lead naturally to equations that have structure like

$$3 = \frac{10}{1 + x}.$$

Solve the equation $3 = \frac{10}{1+x}$ for the exact value of x.

(d) Suppose that $y = a + be^{-kt}$. Solve for t in terms of $a, b, k,$ and y. What does this new equation represent?

Activity 3.6.2. A can of soda is initially at room temperature, $72.3°$ Fahrenheit, and at time $t = 0$ is placed in a refrigerator set at $37.7°$. In addition, we know that after 30 minutes, the soda's temperature has dropped to $59.5°$. Let $F(t)$ represent the temperature of the soda in degrees Fahrenheit at time t in minutes.

(a) Use algebraic reasoning and your understanding of the physical situation to determine the exact values of a, c, and k in the model $F(t) = ae^{-kt} + c$. Write at least one careful sentence to explain your thinking.

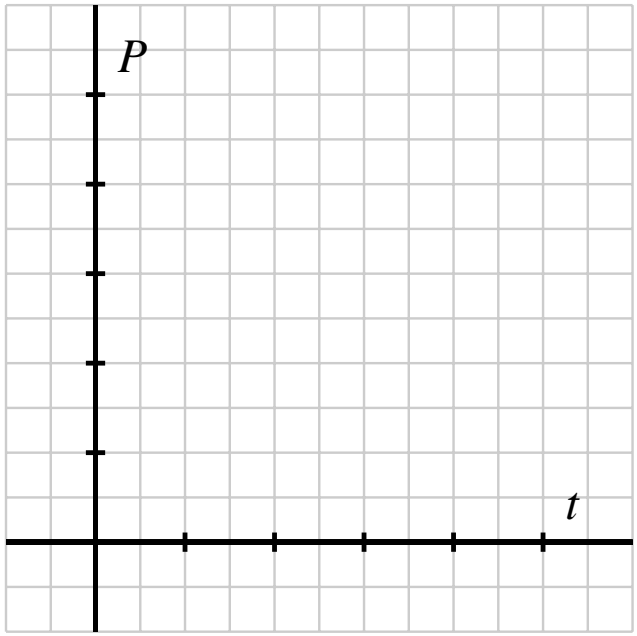
(b) Determine the exact time the object's temperature is $42.4°$. Clearly show your algebraic work and thinking.

(c) In *Desmos*, enter the values you found for a, c, and k in order to define the function F. Then, use *Desmos* to find the average rate of change of F on the interval $[25, 30]$. What is the meaning (with units) of this value?

(d) If everything stayed the same except the value of $F(0)$, and instead $F(0) = 65$, would the value of k be larger or smaller? Why?

Activity 3.6.3. In *Desmos*, define $P(t) = \frac{A}{1+Me^{-kt}}$ and accept sliders for A, M, and k. Set the slider ranges for these parameters as follows: $0.01 \leq A \leq 10$; $0.01 \leq M \leq 10$; $0.01 \leq k \leq 5$.

(a) Sketch a typical graph of $P(t)$ on the axes provided and write several sentences to explain the effects of A, M, and k on the graph of P.



(b) On a typical logistic graph, where does it appear that the population is growing most rapidly? How is this value connected to the carrying capacity, A?

(c) How does the function $1 + Me^{-kt}$ behave as t decreases without bound? What is the algebraic reason that this occurs?

(d) Use your *Desmos* worksheet to find a logistic function P that has the following properties: $P(0) = 2$, $P(2) = 4$, and $P(t)$ approaches 9 as t increases without bound. What are the approximate values of A, M, and k that make the function P fit these criteria?

Activity 3.6.4. Suppose that a population of animals (measured in thousands) that lives on an island is known to grow according to the logistic model, where t is measured in years. We know the following information: $P(0) = 2.45$, $P(3) = 4.52$, and as t increases without bound, $P(t)$ approaches 11.7.

(a) Determine the exact values of A, M, and k in the logistic model

$$P(t) = \frac{A}{1 + Me^{-kt}}.$$

Clearly show your algebraic work and thinking.

(b) Plot your model from (a) and check that its values match the desired characteristics. Then, compute the average rate of change of P on the intervals $[0, 2]$, $[2, 4]$, $[4, 6]$, and $[6, 8]$. What is the meaning (with units) of the values you've found? How is the population growing on these intervals?

(c) Find the exact time value when the population will be 10 (thousand). Show your algebraic work and thinking.

CHAPTER 4

Trigonometry

4.1 Right triangles

Preview Activity 4.1.1. For each of the following situations, sketch a right triangle that satisfies the given conditions, and then either determine the requested missing information in the triangle or explain why you don't have enough information to determine it. Assume that all angles are being considered in radian measure.

(a) The length of the other leg of a right triangle with hypotenuse of length 1 and one leg of length $\frac{3}{5}$.

(b) The lengths of the two legs in a right triangle with hypotenuse of length 1 where one of the non-right angles measures $\frac{\pi}{3}$.

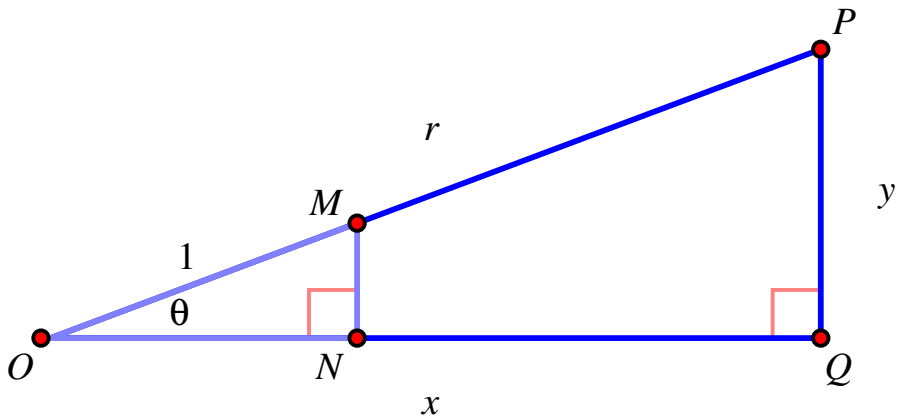
(c) The length of the other leg of a right triangle with hypotenuse of length 7 and one leg of length 6.

(d) The lengths of the two legs in a right triangle with hypotenuse 5 and where one of the non-right angles measures $\frac{\pi}{4}$.

(e) The length of the other leg of a right triangle with hypotenuse of length 1 and one leg of length $\cos(0.7)$.

(f) The measures of the two angles in a right triangle with hypotenuse of length 1 where the two legs have lengths $\cos(1.1)$ and $\sin(1.1)$, respectively.

Activity 4.1.2. Consider right triangle OPQ given in the following figure, and assume that the length of the hypotenuse is $OP = r$ for some constant $r > 1$. Let point M lie on $\overline{OP}$ (the line segment between O and P) in such a way that $OM = 1$, and let point N lie on $\overline{OQ}$ so that $\angle ONM$ is a right angle, as pictured. In addition, assume that point O corresponds to $(0, 0)$, point Q to $(x, 0)$, and point P to (x, y) so that $OQ = x$ and $PQ = y$. Finally, let θ be the measure of $\angle POQ$.



(a) Explain why $\triangle OPQ$ and $\triangle OMN$ are similar triangles.

(b) What is the value of the ratio $\frac{OP}{OM}$? What does this tell you about the ratios $\frac{OQ}{ON}$ and $\frac{PQ}{MN}$?

(c) What is the value of ON in terms of θ? What is the value of MN in terms of θ?

(d) Use your conclusions in (b) and (c) to express the values of x and y in terms of r and θ.

Activity 4.1.3. In each of the following scenarios involving a right triangle, determine the exact values of as many of the remaining side lengths and angle measures (in radians) that you can. If there are quantities that you cannot determine, explain why. For every prompt, draw a labeled diagram of the situation.

(a) A right triangle with hypotenuse 7 and one non-right angle of measure $\frac{\pi}{7}$.

(b) A right triangle with non-right angle α that satisfies $\sin(\alpha) = \frac{3}{5}$.

(c) A right triangle where one of the non-right angles has measure 1.2 and the hypotenuse has length 2.7.

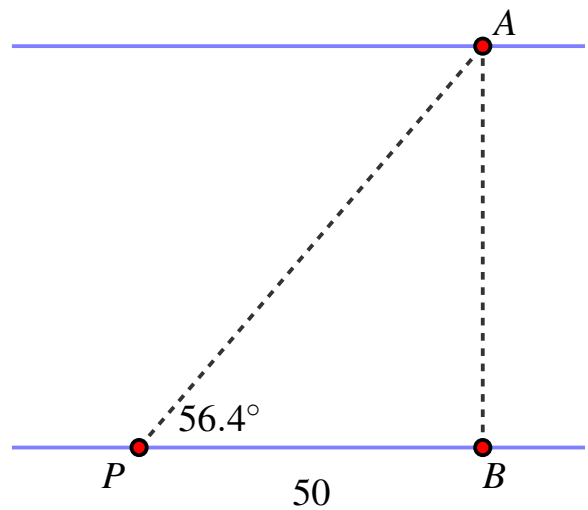
(d) A right triangle with hypotenuse 13 and one leg of length 6.5.

(e) A right triangle with legs of length 5 and 12.

(f) A right triangle where one of the non-right angles has measure $\frac{\pi}{5}$ and the leg opposite this angle has length 4.

Activity 4.1.4.

We want to determine the distance between two points A and B that are directly across from one another on opposite sides of a river, as pictured in the given diagram. We mark the locations of those points and walk 50 meters downstream from B to point P and use a sextant to measure $\angle BPA$. If the measure of $\angle BPA$ is $56.4°$, how wide is the river? What other information about the situation can you determine?



4.2 The Tangent Function

Preview Activity 4.2.1. Through the following questions, we work to understand the special values and overall behavior of the tangent function.

(a) Without using computational device, find the exact value of $\tan(t)$ at the following values: $t = \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{5\pi}{6}$.

(b) Why is $\tan\left(\frac{\pi}{2}\right)$ not defined? What are three other input values x for which $\tan(x)$ is not defined?

(c) Point your browser to http://gvsu.edu/s/0yO (`gvsu.edu/s/0y0`) ("zero-y-Oh") to find a *Desmos* worksheet with data from the tangent function already input, denoted in *Desmos* by the function $T(x) = \tan(x)$. Click on several of the orange points to compare your exact values in (a) with the decimal values given by *Desmos*. Add one entry to the table: $x = \frac{11\pi}{24}$, $y = T(\frac{11\pi}{24})$. At about what coordinates does this point lie? What are the approximate values of $\sin(\frac{11\pi}{24})$ and $\cos(\frac{11\pi}{24})$? Why is the value of $\tan(\frac{11\pi}{24})$ so large relative to the other values of $\tan(x)$ in the table?

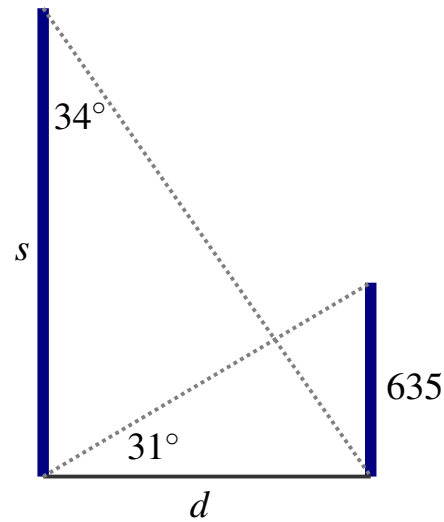
(d) At the top of the input lists on the left side of the *Desmos* worksheet, click the circle to highlight the function $T(x) = \tan(x)$ and thus show its plot along with the data points in orange. Use the plot and your work above to answer the following important questions about the tangent function:

- What is the domain of $y = \tan(x)$?
- What is the period of $y = \tan(x)$?
- What is the range of $y = \tan(x)$?

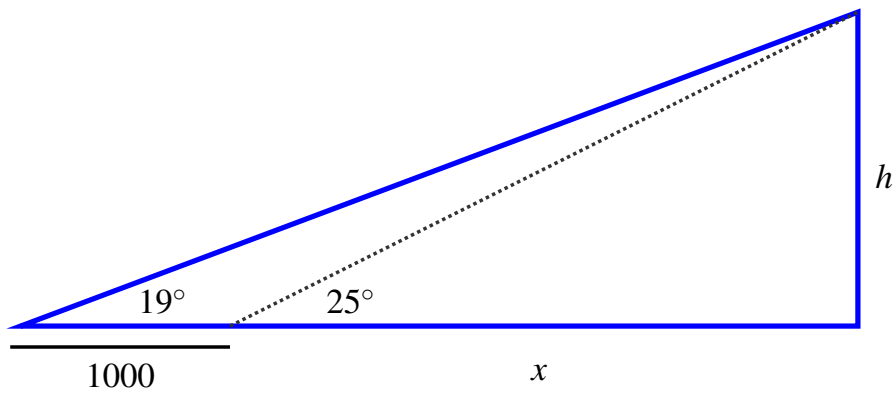
Activity 4.2.2. The top of a 225 foot tower is to be anchored by four cables that each make an angle of $32.5°$ with the ground. How long do the cables have to be and how far from the base of the tower must they be anchored?

Activity 4.2.3.

Supertall high rises have changed the Manhattan skyline. These skyscrapers are known for their small footprint in proportion to their height, with their ratio of width to height at most $1:10$, and some as extreme as $1:24$. Suppose that a relatively short supertall has been built to a height of 635 feet, as pictured in the given figure, and that a second supertall is built nearby. Given the two angles that are computed from the new building, how tall, s, is the new building, and how far apart, d, are the two towers?



Activity 4.2.4. Surveyors are trying to determine the height of a hill relative to sea level. First, they choose a point to take an initial measurement with a sextant that shows the angle of elevation from the ground to the peak of the hill is $19°$. Next, they move 1000 feet closer to the hill, staying at the same elevation relative to sea level, and find that the angle of elevation has increased to $25°$, as pictured in the given diagram. We let h represent the height of the hill relative to the two measurements, and x represent the distance from the second measurement location to the "center" of the hill that lies directly under the peak.



(a) Using the right triangle with the $25°$ angle, find an equation that relates x and h.

(b) Using the right triangle with the $19°$ angle, find a second equation that relates x and h.

(c) Our work in (a) and (b) results in a system of two equations in the two unknowns x and h. Solve each of the two equations for h and then substitute appropriately in order to find a single equation in the variable x.

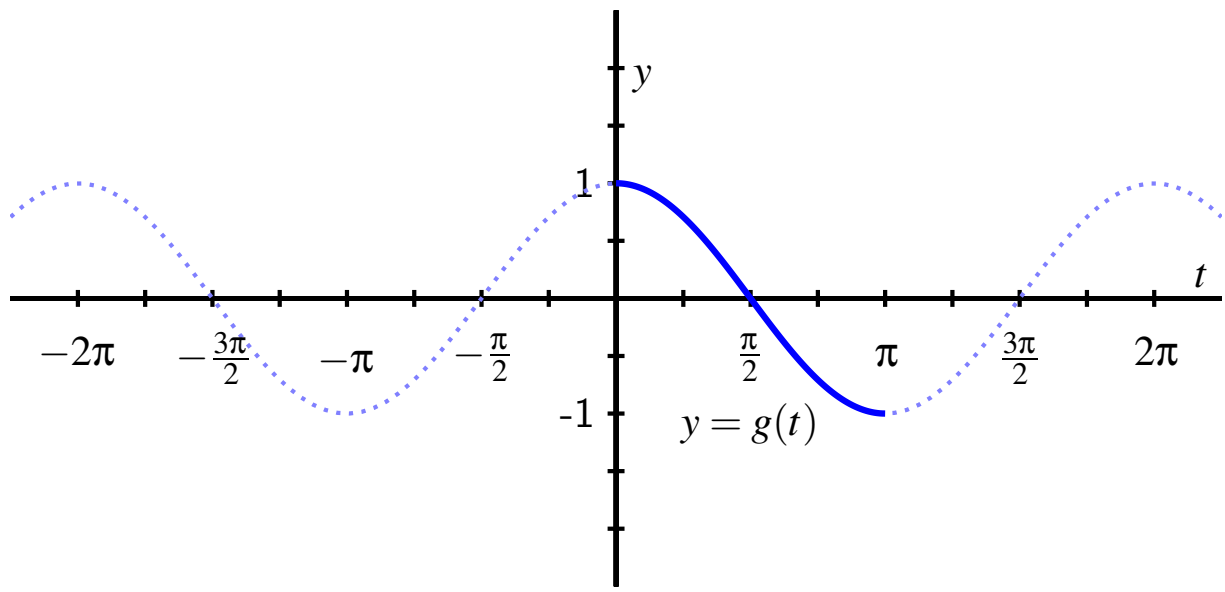
(d) Solve the equation from (c) to find the exact value of x and determine an approximate value accurate to 3 decimal places.

(e) Use your preceding work to solve for h exactly, plus determine an estimate accurate to 3 decimal places.

(f) If the surveyors' initial measurements were taken from an elevation of 78 feet above sea level, how high above sea level is the peak of the hill?

4.3 Inverses of trigonometric functions

Preview Activity 4.3.1. Consider the plot of the standard cosine function in the following figure along with the emphasized portion of the graph on $[0, \pi]$.



Let g be the function whose domain is $0 \leq t \leq \pi$ and whose outputs are determined by the rule $g(t) = \cos(t)$. *Note well*: g is defined in terms of the cosine function, but because it has a different domain, it is *not* the cosine function.

(a) What is the domain of g?

(b) What is the range of g?

(c) Does g pass the horizontal line test? Why or why not?

(d) Explain why g has an inverse function, g^{-1}, and state the domain and range of g^{-1}.

(e) We know that $g(\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$. What is the exact value of $g^{-1}(\frac{\sqrt{2}}{2})$? How about the exact value of $g^{-1}(-\frac{\sqrt{2}}{2})$?

(f) Determine the exact values of $g^{-1}(-\frac{1}{2})$, $g^{-1}(\frac{\sqrt{3}}{2})$, $g^{-1}(0)$, and $g^{-1}(-1)$. Use proper notation to label your results.

Activity 4.3.2. Use the special points on the unit circle (see, for instance, the start of Section 2.3) to determine the exact values of each of the following numerical expressions. Do so without using a computational device.

(a) $\arccos(\frac{1}{2})$

(b) $\arccos(\frac{\sqrt{2}}{2})$

(c) $\arccos(\frac{\sqrt{3}}{2})$

(d) $\arccos(-\frac{1}{2})$

(e) $\arccos(-\frac{\sqrt{2}}{2})$

(f) $\arccos(-\frac{\sqrt{3}}{2})$

(g) $\arccos(-1)$

(h) $\arccos(0)$

(i) $\cos(\arccos(-\frac{1}{2}))$

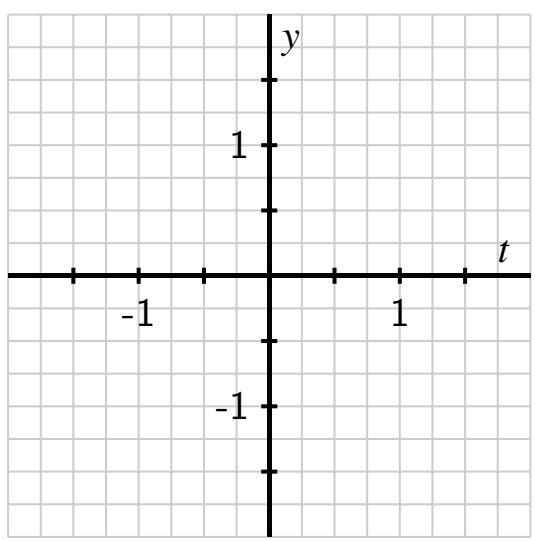
(j) $\arccos(\cos(\frac{7\pi}{6}))$

Activity 4.3.3. The goal of this activity is to understand key properties of the arcsine function in a way similar to our recent discussion of the arccosine function.

(a) Using the definition of the arcsine function, what are the domain and range of the arcsine function?

(b) Determine the following values exactly: $\arcsin(-1)$, $\arcsin(-\frac{\sqrt{2}}{2})$, $\arcsin(0)$, $\arcsin(\frac{1}{2})$, and $\arcsin(\frac{\sqrt{3}}{2})$.

(c) On the axes provided, sketch a careful plot of the restricted sine function on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ along with its corresponding inverse, the arcsine function. Label at least three points on each curve so that each point on the sine graph corresponds to a point on the arcsine graph. In addition, sketch the line $y = t$ to demonstrate how the graphs are reflections of one another across this line.



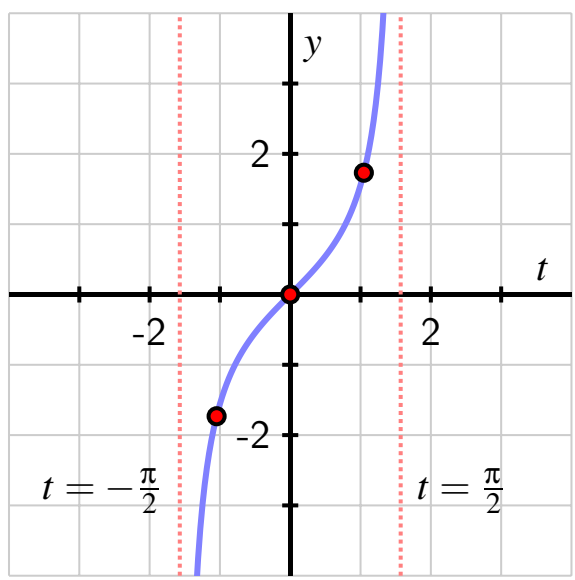
(d) True or false: $\arcsin(\sin(5\pi)) = 5\pi$. Write a complete sentence to explain your reasoning.

Activity 4.3.4. The goal of this activity is to understand key properties of the arctangent function.

(a) Using the definition of the arctangent function, what are the domain and range of the arctangent function?

(b) Determine the following values exactly: $\arctan(-\sqrt{3})$, $\arctan(-1)$, $\arctan(0)$, and $\arctan(\frac{1}{\sqrt{3}})$.

(c) A plot of the restricted tangent function on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ is provided in the following figure. Sketch its corresponding inverse function, the arctangent function, on the same axes. Label at least three points on each curve so that each point on the tangent graph corresponds to a point on the arctangent graph. In addition, sketch the line $y = t$ to demonstrate how the graphs are reflections of one another across this line.



(d) Complete the following sentence: "as t increases without bound, $\arctan(t)$ ...".

4.4 Finding Angles

Preview Activity 4.4.1. Consider a right triangle that has one leg of length 3 and another leg of length $\sqrt{3}$. Let θ be the angle that lies opposite the shorter leg.

(a) Sketch a labeled picture of the triangle.

(b) What is the exact length of the triangle's hypotenuse?

(c) What is the exact value of $\sin(\theta)$?

(d) Rewrite your equation from (c) using the arcsine function in the form $\arcsin(\square) = \Delta$, where $\square$ and Δ are numerical values.

(e) What special angle from the unit circle is θ?

Activity 4.4.2. For each of the following different scenarios, draw a picture of the situation and use inverse trigonometric functions appropriately to determine the missing information both exactly and approximately.

(a) Consider a right triangle with legs of length 11 and 13. What are the measures (in radians) of the non-right angles and what is the length of the hypotenuse?

(b) Consider an angle α in standard position (vertex at the origin, one side on the positive x-axis) for which we know $\cos(\alpha) = -\frac{1}{2}$ and α lies in quadrant III. What is the measure of α in radians? In addition, what is the value of $\sin(\alpha)$?

(c) Consider an angle β in standard position for which we know $\sin(\beta) = 0.1$ and β lies in quadrant II. What is the measure of β in radians? In addition, what is the value of $\cos(\beta)$?

Activity 4.4.3. A roof is being built with a "7-12 pitch." This means that the roof rises 7 inches vertically for every 12 inches of horizontal span; in other words, the slope of the roof is $\frac{7}{12}$. What is the exact measure (in degrees) of the angle the roof makes with the horizontal? What is the approximate measure? What are the exact and approximate measures of the angle at the peak of the roof (made by the front and back portions of the roof that meet to form the ridge)?

Activity 4.4.4. On a baseball diamond (which is a square with 90-foot sides), the third baseman fields the ball right on the line from third base to home plate and 10 feet away from third base (towards home plate). When he throws the ball to first base, what angle (in degrees) does the line the ball travels make with the first base line? What angle does it make with the third base line? Draw a well-labeled diagram to support your thinking.
What angles arise if he throws the ball to second base instead?

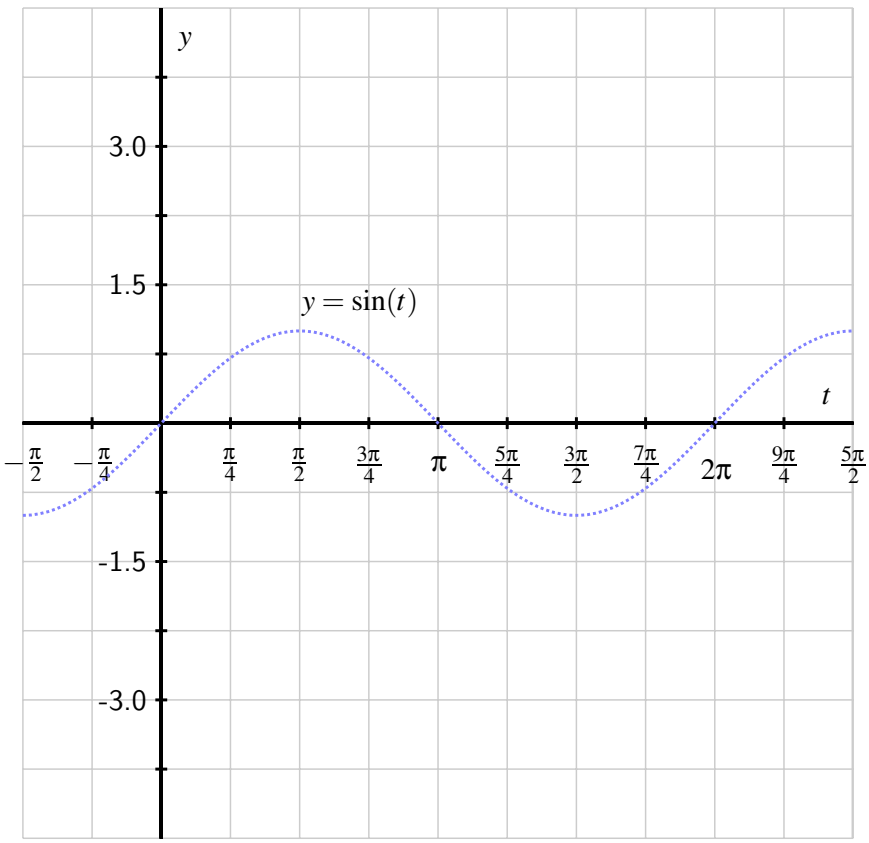
Activity 4.4.5. A camera is tracking the launch of a SpaceX rocket. The camera is located $4000'$ from the rocket's launching pad, and the camera angle changes in order to keep the rocket in focus. At what angle θ (in radians) is the camera tilted when the rocket is $3000'$ off the ground? Answer both exactly and approximately.
Now, rather than considering the rocket at a fixed height of $3000'$, let its height vary and call the rocket's height h. Determine the camera's angle, θ as a function of h, and compute the average rate of change of θ on the intervals $[3000, 3500]$, $[5000, 5500]$, and $[7000, 7500]$. What do you observe about how the camera angle is changing?

4.5 Other Trigonometric Functions and Identities

Preview Activity 4.5.1. Consider a right triangle with hypotenuse of length 61 and one leg of length 11. Let α be the angle opposite the side of length 11. Find the exact length of the other leg and then determine the value of each of the six trigonometric functions evaluated at α. In addition, what are the exact and approximate measures of the two non-right angles in the triangle?

Activity 4.5.2. Suppose that β is an angle in standard position with its terminal side in quadrant II and you know that $\sec(\beta) = -2$. Without using a computational device in any way, determine the exact values of the other five trigonometric functions evaluated at β.

Activity 4.5.3. In this activity, we develop the standard properties of the cosecant function, $q(t) = \csc(t)$.



(a) Complete the tables below to determine the exact values of the cosecant function at the special points on the unit circle. Enter "u" for any value at which $q(t) = \csc(t)$ is undefined.

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\sin(t)$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\csc(t)$									

t	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
$\sin(t)$	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{1}{2}$	0
$\csc(t)$								

(b) In which quadrants is $q(t) = \csc(t)$ positive? negative?

(c) At what t-values does $q(t) = \csc(t)$ have a vertical asymptote? Why?

(d) What is the domain of the cosecant function? What is its range?

(e) Sketch an accurate, labeled graph of $q(t) = \csc(t)$ on the figure given at the start of this activity, including the special points that come from the unit circle.

(f) What is the period of the cosecant function?

Activity 4.5.4. In this activity, we develop the standard properties of the cotangent function, $r(t) = \cot(t)$.

(a) Complete the following tables to determine the exact values of the cotangent function at the special points on the unit circle. Enter "u" for any value at which $r(t) = \cot(t)$ is undefined.

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
$\sin(t)$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\cos(t)$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan(t)$	0	$\frac{1}{\sqrt{3}}$	1	$\frac{3}{\sqrt{3}}$	u	$-\frac{3}{\sqrt{3}}$	-1	$-\frac{1}{\sqrt{3}}$	0
$\cot(t)$									

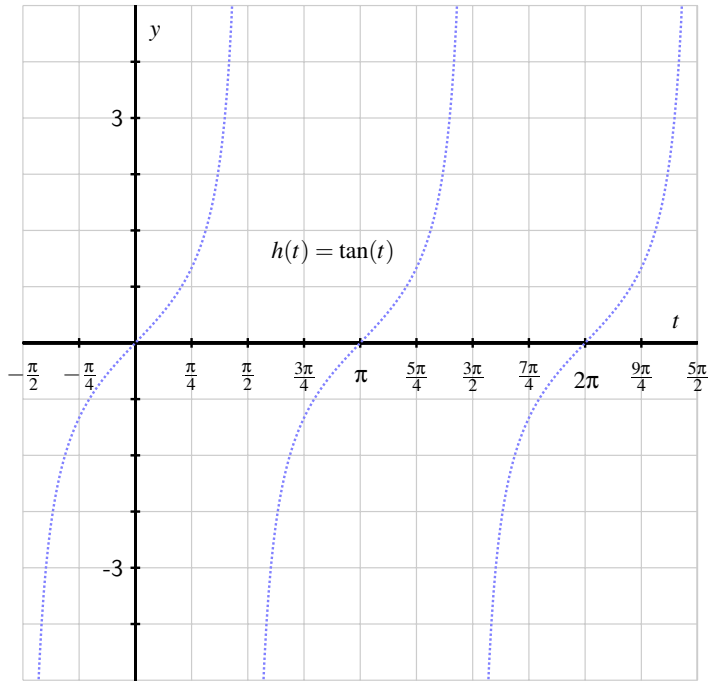
t	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
$\sin(t)$	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{1}{2}$	0
$\cos(t)$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\tan(t)$	$\frac{1}{\sqrt{3}}$	1	$\frac{3}{\sqrt{3}}$	u	$-\frac{3}{\sqrt{3}}$	-1	$-\frac{1}{\sqrt{3}}$	0
$\cot(t)$								

(b) In which quadrants is $r(t) = \cot(t)$ positive? negative?

(c) At what t-values does $r(t) = \cot(t)$ have a vertical asymptote? Why?

(d) What is the domain of the cotangent function? What is its range?

(e) Sketch an accurate, labeled graph of $r(t) = \cot(t)$ on the axes provided below, including the special points that come from the unit circle.



(f) On intervals where the function is defined at every point in the interval, is $r(t) = \cot(t)$ always increasing, always decreasing, or neither?

(g) What is the period of the cotangent function?

(h) How would you describe the relationship between the graphs of the tangent and cotangent functions?

Activity 4.5.5. In this activity, we investigate how a sum of two angles identity for the sine function helps us gain a different perspective on the average rate of change of the sine function.
Recall that for any function f on an interval $[a, a+h]$, its average rate of change is

$$AV_{[a,a+h]} = \frac{f(a+h) - f(a)}{h}.$$

(a) Let $f(x) = \sin(x)$. Use the definition of $AV_{[a,a+h]}$ to write an expression for the average rate of change of the sine function on the interval $[a, a+h]$.

(b) Apply the sum of two angles identity for the sine function, $\sin(\alpha + \beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$, to the expression $\sin(a+h)$.

(c) Explain why your work in (a) and (b) together with some algebra shows that

$$AV_{[a,a+h]} = \sin(a) \cdot \frac{\cos(h) - 1}{h} - \cos(a) \cdot \frac{\sin(h)}{h}.$$

(d) In calculus, we move from *average* rate of change to *instantaneous* rate of change by letting h approach 0 in the expression for average rate of change. Using a computational device in radian mode, investigate the behavior of

$$\frac{\cos(h) - 1}{h}$$

as h gets close to 0. What happens? Similarly, how does $\frac{\sin(h)}{h}$ behave for small values of h? What does this tell us about $AV_{[a,a+h]}$ for the sine function as h approaches 0?

CHAPTER 5

Polynomial and Rational Functions

5.1 Infinity, limits, and power functions

Preview Activity 5.1.1. Complete each of the following statements with an appropriate number or the symbols ∞ or $-\infty$. Do your best to do so *without* using a graphing utility; instead use your understanding of the function's graph.

(a) As $t \to \infty$, $e^{-t} \to$_____.

(b) As $t \to \infty$, $\ln(t) \to$_____.

(c) As $t \to \infty$, $e^{t} \to$_____.

(d) As $t \to 0^+$, $e^{-t} \to$_____. (When we write $t \to 0^+$, this means that we are letting t get closer and closer to 0, but only allowing t to take on positive values.)

(e) As $t \to \infty$, $35 + 53e^{-0.025t} \to$_____.

(f) As $t \to \frac{\pi}{2}^-$, $\tan(t) \to$_____. (When we write $t \to \frac{\pi}{2}^-$, this means that we are letting t get closer and closer to $\frac{\pi}{2}^-$, but only allowing t to take on values that lie to the left of $\frac{\pi}{2}$.)

(g) As $t \to \frac{\pi}{2}^+$, $\tan(t) \to$_____. (When we write $t \to \frac{\pi}{2}^+$, this means that we are letting t get closer and closer to $\frac{\pi}{2}^+$, but only allowing t to take on values that lie to the right of $\frac{\pi}{2}$.)

Activity 5.1.2. Complete the following table by entering "∞,""$-\infty$,""0," or "no limit" to identify how the function behaves as either x increases or decreases without bound. As much as possible, work to decide the behavior *without* using a graphing utility.

$f(x)$	$\lim_{x\to\infty} f(x)$	$\lim_{x\to-\infty} f(x)$
e^x		
e^{-x}		
$\ln(x)$		
x		
x^2		
x^3		
x^4		
$\frac{1}{x}$		
$\frac{1}{x^2}$		
$\sin(x)$		

Activity 5.1.3. Point your browser to the *Desmos* worksheet at http://gvsu.edu/s/0zu (gvsu.edu/s/0zu). In what follows, we explore the behavior of power functions of the form $y = x^n$ where $n \geq 1$.

(a) Press the "play" button next to the slider labeled "n." Watch at least two loops of the animation and then discuss the trends that you observe. Write a careful sentence each for at least two different trends.

(b) Click the icons next to each of the following 8 functions so that you can see all of $y = x$, $y = x^2$, ..., $y = x^8$ graphed at once. On the interval $0 < x < 1$, how do the graphs of x^a and x^b compare if $a < b$?

(c) Uncheck the icons on each of the 8 functions to hide their graphs. Click the settings icon to change the domain settings for the axes, and change them to $-10 \leq x \leq 10$ and $-10,000 \leq y \leq 10,000$. Play the animation through twice and then discuss the trends that you observe. Write a careful sentence each for at least two different trends.

(d) Click the icons next to each of the following 8 functions so that you can see all of $y = x$, $y = x^2$, ..., $y = x^8$ graphed at once. On the interval $x > 1$, how do the graphs of x^a and x^b compare if $a < b$?

Activity 5.1.4. Point your browser to the *Desmos* worksheet at http://gvsu.edu/s/0zv (`gvsu.edu/s/0zv`). In what follows, we explore the behavior of power functions $y = x^n$ where $n \leq -1$.

(a) Press the "play" button next to the slider labeled "n." Watch two loops of the animation and then discuss the trends that you observe. Write a careful sentence each for at least two different trends.

(b) Click the icons next to each of the following 8 functions so that you can see all of $y = x^{-1}$, $y = x^{-2}$, ..., $y = x^{-8}$ graphed at once. On the interval $1 < x$, how do the functions x^a and x^b compare if $a < b$? (Be careful with negative numbers here: e.g., $-3 < -2$.)

(c) How do your answers change on the interval $0 < x < 1$?

(d) Uncheck the icons on each of the 8 functions to hide their graphs. Click the settings icon to change the domain settings for the axes, and change them to $-10 \leq x \leq 10$ and $-10,000 \leq y \leq 10,000$. Play the animation through twice and then discuss the trends that you observe. Write a careful sentence each for at least two different trends.

(e) Explain why $\lim_{x\to\infty} \frac{1}{x^n} = 0$ for any choice of $n = 1, 2, \ldots$.

5.2 Polynomials

Preview Activity 5.2.1. Point your browser to the *Desmos* worksheet at http://gvsu.edu/s/0zy (`gvsu.edu/s/0zy`). There you'll find a degree 4 polynomial of the form $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$, where $a_0, \ldots, a_4$ are set up as sliders. In the questions that follow, you'll experiment with different values of $a_0, \ldots, a_4$ to investigate different possible behaviors in a degree 4 polynomial. Note that we require $a_4 \neq 0$ in order to ensure p is a degree 4 polynomial.

(a) What is the largest number of distinct points at which $p(x)$ can cross the x-axis?

Recall from the definition of a polynoimal function what we mean by a **zero** of the polynomial. Give examples of values for $a_0, \ldots, a_4$ that lead to that largest number of zeros for $p(x)$.

(b) What other numbers of zeros are possible for $p(x)$? Said differently, can you get each possible number of fewer zeros than the largest number that you found in (a)? Why or why not?

(c) We say that a function has a **turning point** if the function changes from decreasing to increasing or increasing to decreasing at the point. For example, any quadratic function has a turning point at its vertex.

What is the largest number of turning points that $p(x)$ (the function in the *Desmos* worksheet) can have? Experiment with the sliders, and give examples of values for $a_0, \ldots, a_4$ that lead to that largest number of turning points for $p(x)$.

(d) What other numbers of turning points are possible for $p(x)$? Can it have no turning points? Just one? Exactly two? Experiment and explain.

(e) What long-range behavior is possible for $p(x)$? Said differently, what are the possible results for $\lim_{x\to-\infty} p(x)$ and $\lim_{x\to\infty} p(x)$?

(f) What happens when we plot $y = a_4x^4$ in *Desmos* and compare $p(x)$ and a_4x^4? How do they look when we zoom out? (Experiment with different values of each of the sliders, too.)

Activity 5.2.2. By experimenting with coefficients in *Desmos*, find a formula for a polynomial function that has the stated properties, or explain why no such polynomial exists. If you enter `p(x)=a+bx+cx^2+dx^3+fx^4+gx^5` in *Desmos*, you'll get prompted to add sliders that make it easy to explore a degree 5 polynomial. (We skip using `e` as one of the constants since *Desmos* reserves `e` as the Euler constant.)

(a) A polynomial p of degree 5 with exactly 3 real zeros, 4 turning points, and such that $\lim_{x\to-\infty} p(x) = +\infty$ and $\lim_{x\to\infty} p(x) = -\infty$.

(b) A polynomial p of degree 4 with exactly 4 real zeros, 3 turning points, and such that $\lim_{x\to-\infty} p(x) = +\infty$ and $\lim_{x\to\infty} p(x) = -\infty$.

(c) A polynomial p of degree 6 with exactly 2 real zeros, 3 turning points, and such that $\lim_{x\to-\infty} p(x) = -\infty$ and $\lim_{x\to\infty} p(x) = -\infty$.

(d) A polynomial p of degree 5 with exactly 5 real zeros, 3 turning points, and such that $\lim_{x\to-\infty} p(x) = +\infty$ and $\lim_{x\to\infty} p(x) = -\infty$.

Activity 5.2.3. Consider the polynomial function given by

$$p(x) = 4692(x + 1520)(x^2 + 10000)(x - 3471)^2(x - 9738).$$

(a) What is the degree of p? How can you tell *without* fully expanding the factored form of the function?

(b) What can you say about the sign of the factor $(x^2 + 10000)$?

(c) What are the zeros of the polynomial p?

(d) Construct a sign chart for p by using the zeros you identified in (c) and then analyzing the sign of each factor of p.

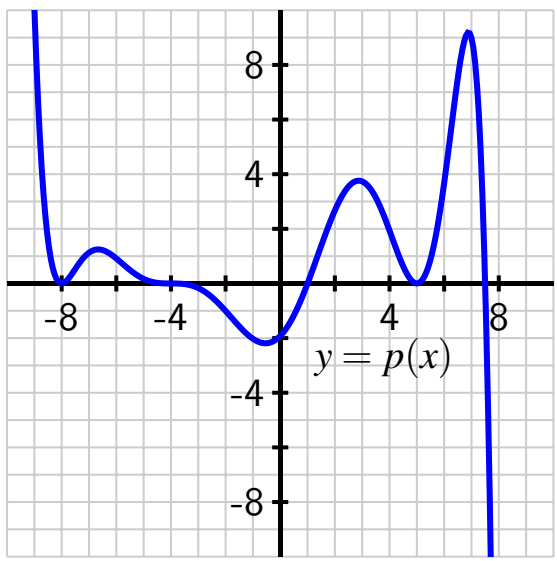
(e) Without using a graphing utility, construct an approximate graph of p that has the zeros of p carefully labeled on the x-axis.

(f) Use a graphing utility to check your earlier work. What is challenging or misleading when using technology to graph p?

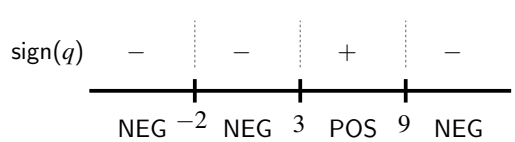
Activity 5.2.4. For each of the following prompts, try to determine a formula for a polynomial that satisfies the given criteria. If no such polynomial exists, explain why.

(a) A polynomial f of degree 10 whose zeros are $x = -12$ (multiplicity 3), $x = -9$ (multiplicity 2), $x = 4$ (multiplicity 4), and $x = 10$ (multiplicity 1), and f satisfies $f(0) = 21$. What can you say about the values of $\lim_{x\to-\infty} f(x)$ and $\lim_{x\to\infty} f(x)$?

(b) A polynomial p of degree 9 that satisfies $p(0) = -2$ and has the graph shown in the following figure. Assume that all of the zeros of p are shown in the figure.



(c) A polynomial q of degree 8 with 3 distinct real zeros (possibly of different multiplicities) such that q has the sign chart in the figure below and satisfies $q(0) = -10$.



(d) A polynomial q of degree 9 with 3 distinct real zeros (possibly of different multiplicities) such that q satisfies the sign chart in part (c) and satisfies $q(0) = -10$.

(e) A polynomial p of degree 11 that satisfies $p(0) = -2$ and p has the graph shown in part (b). Assume that all of the zeros of p are shown in the figure.

5.3 Modeling with polynomial functions

Preview Activity 5.3.1. A piece of cardboard that is 12×18 (each measured in inches) is being made into a box without a top. To do so, squares are cut from each corner of the cardboard and the remaining sides are folded up.

(a) Let x be the side length of the squares being cut from the corners of the cardboard. Draw a labeled diagram that shows the given information and the variable being used.

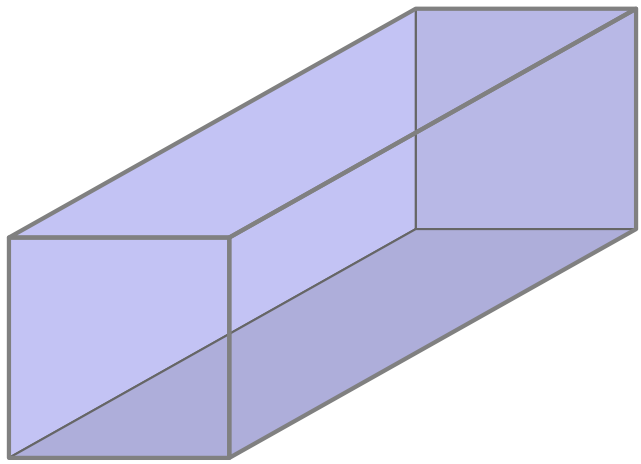
(b) Determine a formula for the function V whose output is the volume of the box that results from a square of size $x \times x$ being cut from each corner of the cardboard.

(c) What familiar kind of function is V?

(d) If we start with a small positive value for x and let that value get larger and larger, what is the first value of $x$we encounter that makes it impossible to remove $x \times x$ squares from the cardboard and still form a box?

(e) What are the zeros of V? What is the domain of the model V in the context of the rectangular box?

Activity 5.3.2. According to a shipping company's regulations, the girth plus the length of a parcel they transport for their lowest rate may not exceed 120 inches, where by *girth* we mean the perimeter of one end.



Suppose that we want to ship a parcel that has a square end of width x and an overall length of y, both measured in inches.

(a) Label the provided picture, using x for the length of each side of the square end, and y for the other edge of the package.

(b) How does the length plus girth of 120 inches result in an equation (often called a *constraint* equation) that relates x and y? Explain, and state the equation.

(c) Solve the equation you found in (b) for one of the variables present.

(d) Hence determine the volume, V, of the package as a function of a single variable.

(e) What is the domain of the function V in the context of the physical setting of this problem? (Hint: neither x nor y can equal 0.)

Activity 5.3.3. Suppose that we want to construct a cylindrical can using 60 square inches of material for the surface of the can. In this context, how does the can's volume depend on the radius we choose?
Let the cylindrical can have base radius r and height h.

(a) Use the formula for the surface area of a cylinder and the given constraint that the can's surface area is 60 square inches to write an equation that connects the radius r and height h.

(b) Solve the equation you found in (a) for h in terms of r.

(c) Recall that the volume of a cylinder is $V = \pi r^2 h$. Use your work in (b) to write V as a function of the single variable r; simplify the formula as much as possible.

(d) What is the domain of the function V in the context of the physical setting of this problem? (Hint: how does the constraint on surface area provide an upper bound for the value of r? Think about the maximum area that can be allocated to the top and bottom of the can.)

Activity 5.3.4. We understand the theoretical rule behind the function $f(t) = \sin(t)$: given an angle t in radians, $\sin(t)$ measures the value of the y-coordinate of the corresponding point on the unit circle. For special values of t, we have determined the exact value of $\sin(t)$. For example, $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. But note that we don't have a *formula* for $\sin(t)$. Instead, we use a button on our calculator or command on our computer to find values like "$\sin(1.35)$." It turns out that a combination of calculus and polynomial functions explains how computers determine values of the sine function.
At http://gvsu.edu/s/0zA (`gvsu.edu/s/0zA`), you'll find a *Desmos* worksheet that has the sine function already defined, along with a sequence of polynomials labeled $T_1(x), T_3(x), T_5(x), T_7(x), \ldots$. You can see these functions' graphs by clicking on their respective icons.

(a) For what values of x does it appear that $\sin(x) \approx T_1(x)$?

(b) For what values of x does it appear that $\sin(x) \approx T_3(x)$?

(c) For what values of x does it appear that $\sin(x) \approx T_5(x)$?

(d) What overall trend do you observe? How good is the approximation generated by $T_{19}(x)$?

(e) In a new *Desmos* worksheet, plot the function $y = \cos(x)$ along with the following functions: $P_2(x) = 1 - \frac{x^2}{2!}$ and $P_4(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$. Based on the patterns with the coefficients in the polynomials approximating $\sin(x)$ and the polynomials P_2 and P_4 here, conjecture formulas for P_6, P_8, and P_{18} and plot them. How well can we approximate $y = \cos(x)$ using polynomials?

5.4 Rational Functions

Preview Activity 5.4.1. A drug company estimates that to produce a new drug, it will cost \$5 million in startup resources, and that once they reach production, each gram of the drug will cost \$2500 to make.

(a) Determine a formula for a function $C(q)$ that models the cost of producing q grams of the drug. What familiar kind of function is C?

(b) The drug company needs to sell the drug at a price of more than \$2500 per gram in order to at least break even. To investigate how they might set prices, they first consider what their *average* cost per gram is. What is the *total* cost of producing 1000 grams? What is the *average* cost per gram to produce 1000 grams?

(c) What is the total cost of producing 10000 grams? What is the average cost per gram to produce 10000 grams?

(d) Our computations in (b) and (c) naturally lead us to define the "average cost per gram" function, $A(q)$, whose output is the average cost of producing q grams of the drug. What is a formula for $A(q)$?

(e) Explain why another formula for A is $A(q) = 2500 + \frac{5000000}{q}$.

(f) What can you say about the long-range behavior of A? What does this behavior mean in the context of the problem?

This activity is based on p. 457ff in *Functions Modeling Change*, by Connally et al.

Activity 5.4.2. Consider the rational function $r(x) = \frac{3x^2-5x+1}{7x^2+2x-11}$.
Observe that the largest power of x that's present in $r(x)$ is x^2. In addition, because of the dominant terms of $3x^2$ in the numerator and $7x^2$ in the denominator, both the numerator and denominator of r increase without bound as x increases without bound. In order to understand the long-range behavior of r, we choose to write the function in a different algebraic form.

(a) Note that we can multiply the formula for r by the form of 1 given by $1 = \frac{\frac{1}{x^2}}{\frac{1}{x^2}}$. Do so, and distribute and simplify as much as possible in both the numerator and denominator to write r in a different algebraic form.

(b) Having rewritten r, we are in a better position to evaluate $\lim_{x\to\infty} r(x)$. Using our work from (a), we have

$$\lim_{x\to\infty} r(x) = \lim_{x\to\infty} \frac{3 - \frac{5}{x} + \frac{1}{x^2}}{7 + \frac{2}{x} - \frac{11}{x^2}}.$$

What is the exact value of this limit and why?

(c) Next, determine

$$\lim_{x\to-\infty} r(x) = \lim_{x\to-\infty} \frac{3 - \frac{5}{x} + \frac{1}{x^2}}{7 + \frac{2}{x} - \frac{11}{x^2}}.$$

(d) Use *Desmos* to plot r on the interval $[-10, 10]$. In addition, plot the horizontal line $y = \frac{3}{7}$. What is the meaning of the limits you found in (b) and (c)?

Activity 5.4.3. Let $s(x) = \frac{3x-5}{7x^2+2x-11}$ and $u(x) = \frac{3x^2-5x+1}{7x+2}$. Note that both the numerator and denominator of each of these rational functions increases without bound as $x \to \infty$, and in addition that x^2 is the highest order term present in each of s and u.

(a) Using a similar algebraic approach to our work in Activity 5.4.2, multiply $s(x)$ by $1 = \frac{\frac{1}{x^2}}{\frac{1}{x^2}}$ and hence evaluate

$$\lim_{x\to\infty} \frac{3x-5}{7x^2+2x-11}.$$

What value do you find?

(b) Plot the function $y = s(x)$ on the interval $[-10, 10]$. What is the graphical meaning of the limit you found in (a)?

(c) Next, use appropriate algebraic work to consider $u(x)$ and evaluate

$$\lim_{x\to\infty} \frac{3x^2-5x+1}{7x+2}.$$

What do you find?

(d) Plot the function $y = u(x)$ on the interval $[-10, 10]$. What is the graphical meaning of the limit you computed in (c)?

Activity 5.4.4. Determine the domain of each of the following functions. In each case, write a sentence to accurately describe the domain.

(a) $f(x) = \dfrac{x^2 - 1}{x^2 + 1}$

(b) $g(x) = \dfrac{x^2 - 1}{x^2 + 3x - 4}$

(c) $h(x) = \dfrac{1}{x} + \dfrac{1}{x - 1} + \dfrac{1}{x - 2}$

(d) $j(x) = \dfrac{(x+5)(x-3)(x+1)(x-4)}{(x+1)(x+3)(x-5)}$

(e) $k(x) = \dfrac{2x^2+7}{3x^3-12x}$

(f) $m(x) = \dfrac{5x^2-45}{7(x-2)(x-3)^2(x^2+9)(x+1)}$

Activity 5.4.5. Suppose that we want to build an open rectangular box (that is, without a top) that holds 15 cubic feet of volume. If we want one side of the base to be twice as long as the other, how does the amount of material required depend on the shorter side of the base? We investigate this question through the following sequence of prompts.

(a) Draw a labeled picture of the box. Let x represent the shorter side of the base and h the height of the box. What is the length of the longer side of the base in terms of x?

(b) Use the given volume constraint to write an equation that relates x and h, and solve the equation for h in terms of x.

(c) Determine a formula for the surface area, S, of the box in terms of x and h.

(d) Using the constraint equation from (b) together with your work in (c), write surface area, S, as a function of the single variable x.

(e) What type of function is S? What is its domain?

(f) Plot the function S using *Desmos*. What appears to be the least amount of material that can be used to construct the desired box that holds 15 cubic feet of volume?

5.5 Key features of rational functions

Preview Activity 5.5.1. Consider the rational function $r(x) = \frac{x^2-1}{x^2-3x-4}$, and let $p(x) = x^2 - 1$ (the numerator of $r(x)$) and $q(x) = x^2 - 3x - 4$ (the denominator of $r(x)$).

(a) Reasoning algebraically, for what values of x is $p(x) = 0$?

(b) Again reasoning algebraically, for what values of x is $q(x) = 0$?

(c) Define $r(x)$ in *Desmos*, and evaluate the function appropriately to find numerical values for the output of r and hence complete the following tables.

x	$r(x)$
4.1	
4.01	
4.001	
3.9	
3.99	
3.999	

x	$r(x)$
1.1	
1.01	
1.001	
0.9	
0.99	
0.999	

x	$r(x)$
−1.1	
−1.01	
−1.001	
−0.9	
−0.99	
−0.999	

(d) Why does r behave the way it does near $x = 4$? Explain by describing the behavior of the numerator and denominator.

(e) Why does r behave the way it does near $x = 1$? Explain by describing the behavior of the numerator and denominator.

(f) Why does r behave the way it does near $x = -1$? Explain by describing the behavior of the numerator and denominator.

(g) Plot r in *Desmos*. Is there anything surprising or misleading about the graph that *Desmos* generates?

Activity 5.5.2. For each of the following rational functions, state the function's domain and determine the locations of all zeros, vertical asymptotes, and holes. Provide clear justification for your work by discussing the zeros of the numerator and denominator, as well as a table of values of the function near any point where you believe the function has a hole. In addition, state the value of the horizontal asymptote of the function or explain why the function has no such asymptote.

(a) $f(x) = \dfrac{x^3 - 6x^2 + 5x}{x^2 - 1}$

(b) $g(x) = \dfrac{11(x^2 + 1)(x - 7)}{23(x - 1)(x^2 + 4)}$

(c) $h(x) = \dfrac{x^2 - 8x + 12}{x^2 - 3x - 18}$

(d) $q(x) = \dfrac{(x-2)(x^2-9)}{(x-3)(x^2+4)}$

(e) $r(x) = \dfrac{19(x-2)(x-3)^2(x+1)}{17(x+1)(x-4)^2(x-5)}$

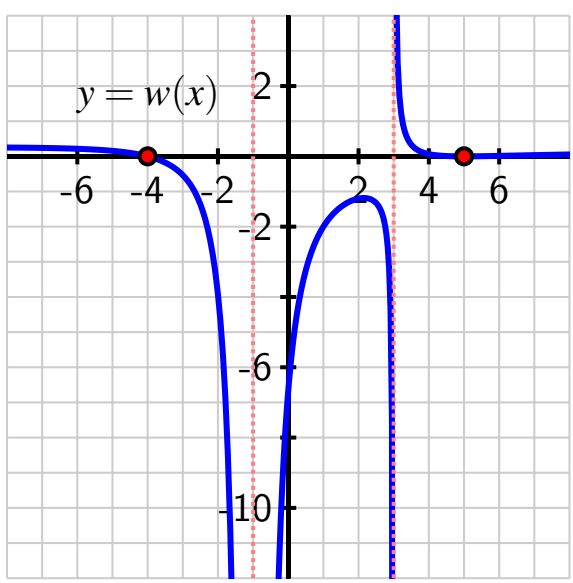
(f) $s(x) = \dfrac{1}{x^2+1}$

Activity 5.5.3. Find a formula for a rational function that meets the stated criteria as given by words, a sign chart, or graph. Write several sentences to justify why your formula matches the specifications.

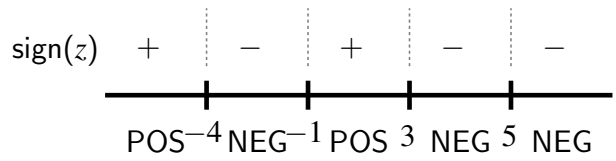
(a) A rational function r such that r has a vertical asymptote at $x = -2$, a zero at $x = 1$, a hole at $x = 5$, and a horizontal asymptote of $y = -3$.

(b) A rational function u whose numerator has degree 3, denominator has degree 3, and that has exactly one vertical asymptote at $x = -4$ and a horizontal asymptote of $y = \frac{3}{7}$.

(c) A rational function w whose formula generates a graph with all of the characteristics shown in the following figure. Assume that $w(5) = 0$ but $w(x) > 0$ for all other x such that $x > 3$.



(d) A rational function z whose formula satisfies the sign chart shown in the following figure, and for which z has no horizontal asymptote and its only vertical asymptotes occur at the middle two values of x noted on the sign chart.



(e) A rational function f that has exactly two holes, two vertical asymptotes, two zeros, and a horizontal asymptote.

Colophon

This book was authored in PreTeXt.